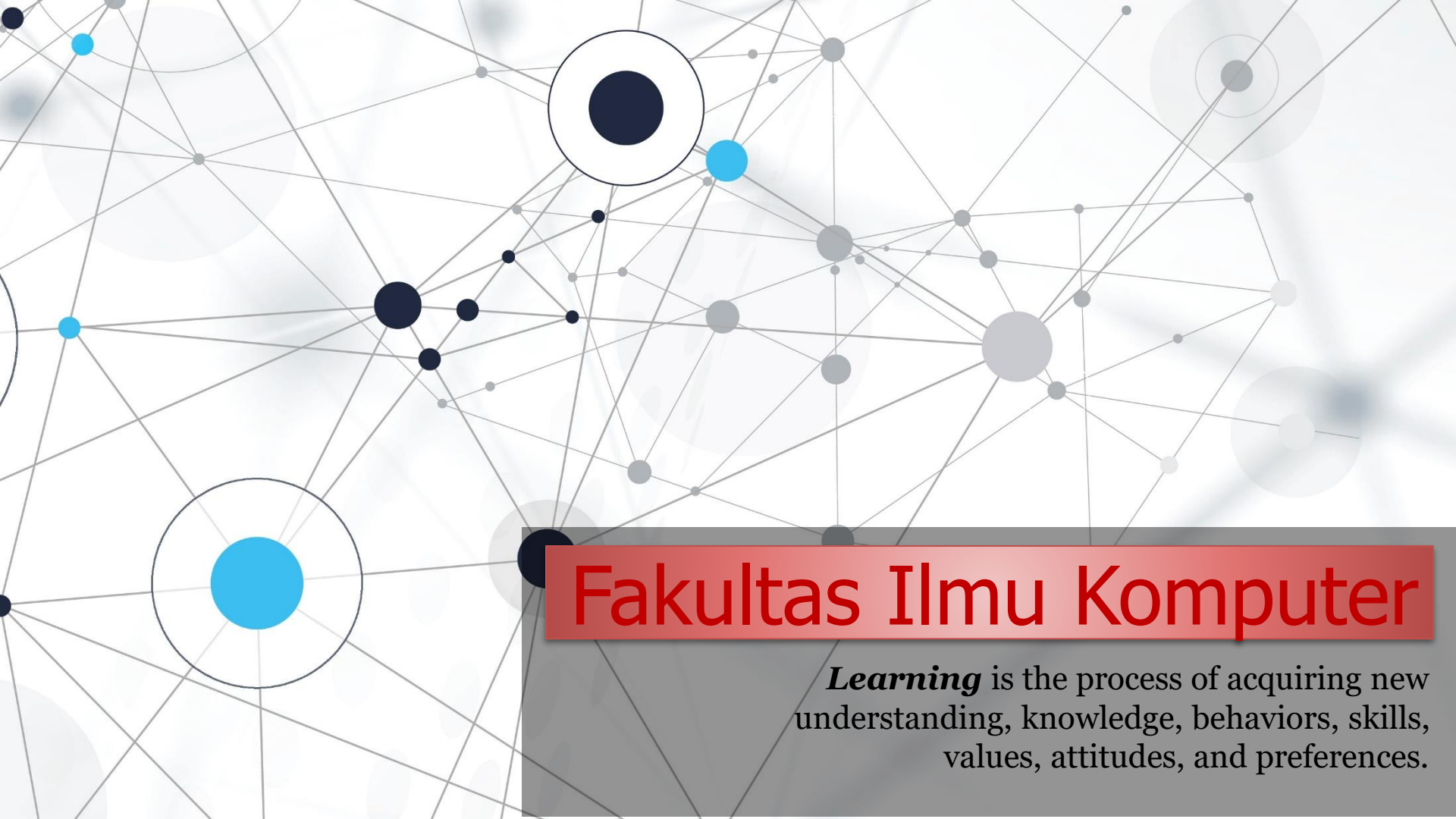




Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Back-end Framework #6 - UI [Bootstrap Open-source Front-end Framework & JQuery]



Welcome to the Back-end Web Development *course*

A skilled Front-End Web Developer not only contributes to front-end development, but can also add significant value to the team and the project as a whole from a back-end/server perspective.

Back End Developer

MUST HAVE SKILLS



Framework Back End

- Django
- Flash
- Rails
- Spring, dll

API

Application Programming Interface

Bahasa Pemrograman

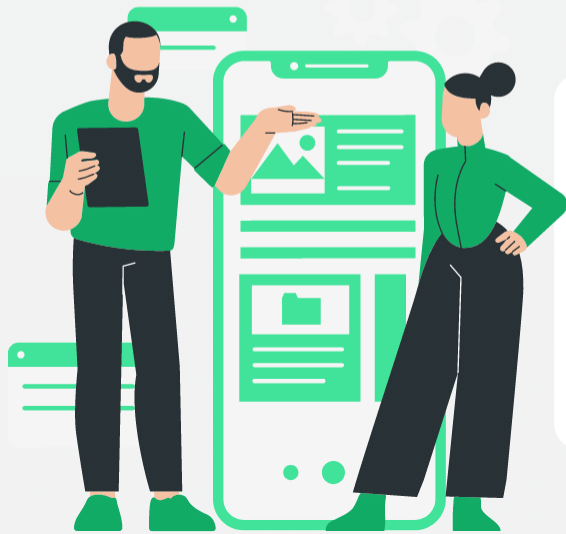
- PHP
- Python
- Ruby
- Javascript

Manajemen Database

- MySQL,
- PostgreSQL,
- noSQL

Pengelolaan Web Server

- Litespeed
- Apache
- Nginx



Tugas dan Tanggung jawab **Back End Developer**

01.

Membuat program yang aman dari hacker/virus

02.

Mendesain struktur data website dan database

03.

Mengembangkan coding dan menemukan error

04.

Update Produk/Aplikasi

05.

Mengembangkan struktur data dasar

06.

Bekerjasama dengan Front-End untuk tampilan produk

07.

Mengatasi masalah pada server

From Back-end/Server Perspective

Front-End Web Developer

Seorang **Front-End Web Developer** adalah *Software Developer* yang bertanggung jawab untuk merancang dan membangun berbagai aplikasi *web* secara **responsif**, **interaktif**, dan juga **user friendly**. Seorang Front-End Web Developer memiliki peranan besar dalam pengembangan aplikasi web karena bertanggung jawab langsung kepada **pengalaman pengguna**.



A skilled Front-End Developer will have a good understanding of what is required from the user side (user-facing). They will understand the design and functionality demands that must be implemented on the *front end*.

Read Source:

<https://www.dicoding.com/learningpaths/22/>

Front-End Web Development



Developer memegang peranan penting di dalam pengembangan sebuah *situs* atau *aplikasi*. Apakah perbedaan *front end* dan *back end*?

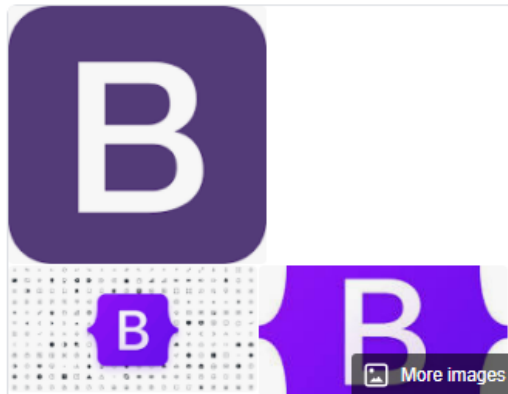
Front-end & Back-end Developer :

Front end developer adalah orang yang berperan mengembangkan tampilan situs atau aplikasi melalui HTML, CSS, dan JavaScript.

Sedangkan, **back end developer** adalah orang yang berperan agar situs atau aplikasi dapat bekerja.

Bootstrap & JQuery

Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development. **jQuery** is a JavaScript library designed to simplify HTML DOM tree traversal and manipulation, as well as event handling, CSS animation, and Ajax.



Bootstrap

Front-end framework

Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development. It contains CSS- and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.

[Wikipedia](#)

Stable release: 5.0.1 / 13 May 2021; 34 days ago

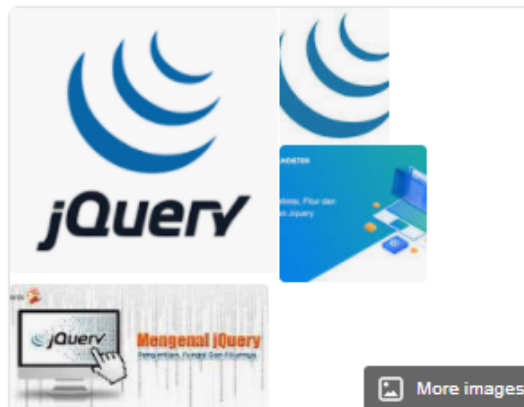
Initial release: August 19, 2011; 9 years ago

Original author(s): Mark Otto, Jacob Thornton

Written in: [HTML](#), [CSS](#), [Less](#) (v3), [Sass](#) (v4) and [JavaScript](#)

Platform: [Web platform](#)

License: [MIT License](#) ([Apache License 2.0](#) prior to 3.1.0)



jQuery

Software

jQuery is a JavaScript library designed to simplify HTML DOM tree traversal and manipulation, as well as event handling, CSS animation, and Ajax. It is free, open-source software using the permissive MIT License. As of May 2019, jQuery is used by 73% of the 10 million most popular websites. [Wikipedia](#)

Stable release: 3.6.0 / (March 2, 2021; 3 months ago)

Initial release: August 26, 2006; 14 years ago

Original author(s): John Resig

Size: 27–274 KB

License: [MIT](#)

Written in: [JavaScript](#)

How to download **Bootstrap** and **jQuery** and add/use them offline in the project?



File/Directory Framework Structure

No	Directory/File	Description (<i>optional</i>)
1	app	app directory is the web application's private directory of this new framework.
2	public	public directory is to store all the public static resources like html , css , image and js files.
3	core	core directory is to store all files for routing and it will store the framework's core classes.
4	model	model directory is for all app's Model classes.
5	view	view directory is for all app's View classes.
6	controller	controller directory is for all app's Controller classes.
7	css	css directory is for Cascading Style Sheets (CSS) files.
8	img	Img directory is for images files.
9	js	js directory is for Javascript files.
10	index.php	<i>index.php</i> file is the “ <i>entry-point</i> ” file, the front controller to the users.

Additional Directories:

template directory stores the template files like *header* file and *footer* file.

config directory stores the app's configuration files.

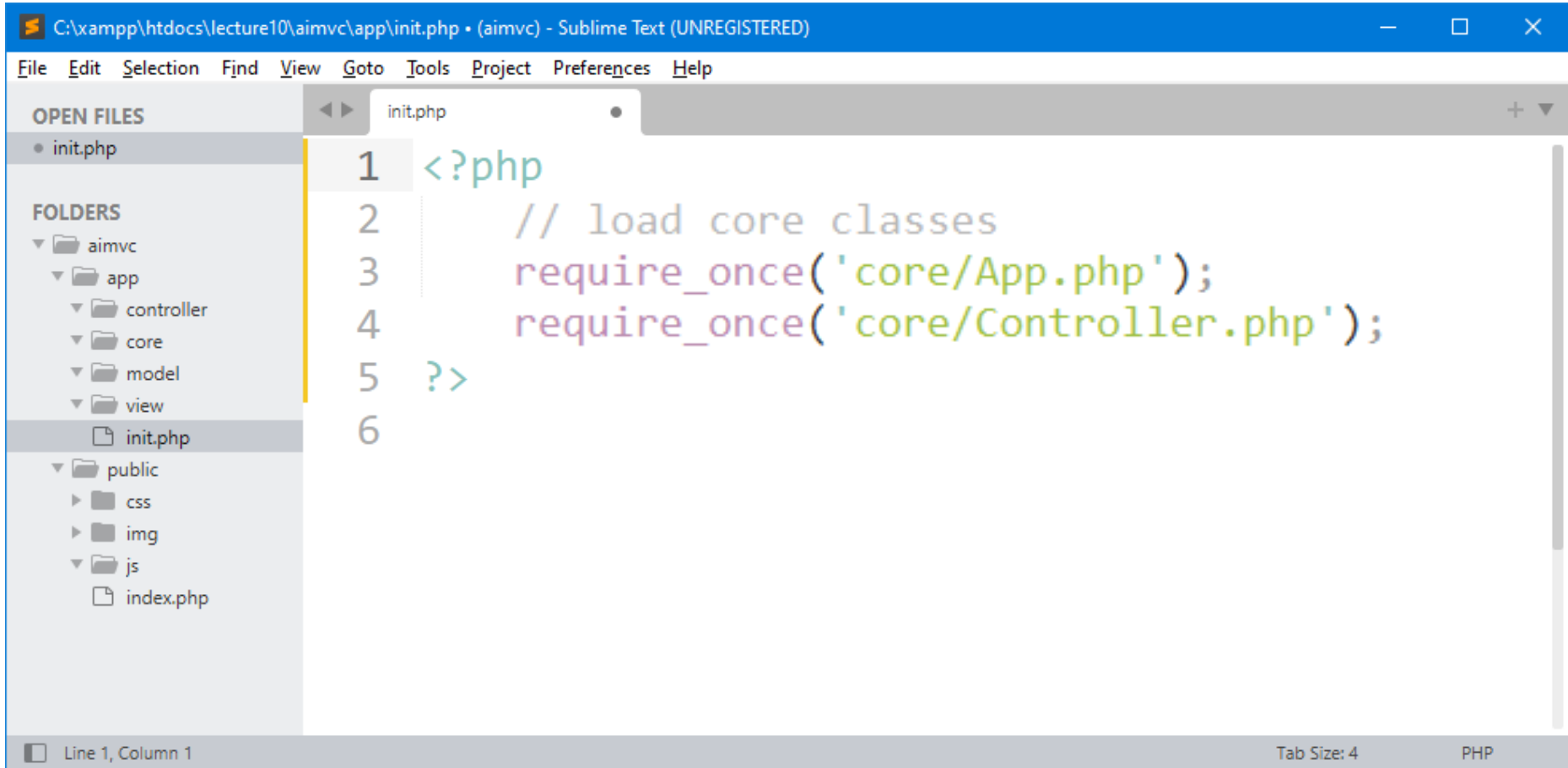
upload directory is for uploaded files, such as uploaded images.

db directory is database related classes, such as database driver classes.

helper directory is help/assistant functions.

library directory is for class libraries.

MVC Assets Directory



The screenshot shows a Sublime Text editor window titled "C:\xampp\htdocs\lecture10\aimvc\app\init.php • (aimvc) - Sublime Text (UNREGISTERED)". The editor displays the contents of the `init.php` file, which is a PHP script that loads core classes. The code is as follows:

```
1 <?php
2     // load core classes
3     require_once('core/App.php');
4     require_once('core/Controller.php');
5 ?>
6
```

The left sidebar shows the "OPEN FILES" and "FOLDERS" panels. The "FOLDERS" panel displays the directory structure of the `aimvc` project:

- aimvc
 - app
 - controller
 - core
 - model
 - view
 - init.php
 - public
 - css
 - img
 - js
 - index.php

The status bar at the bottom indicates "Line 1, Column 1", "Tab Size: 4", and "PHP".

The **assets directory** is more like a files system and provides more freedom to put any file you would like in the directory. Here, we used **Public Directory** as a assets directory which store some public static resources like *css*, *image* and *js* files. The public directory contains the *index.php* file, which is the entry point for all requests entering the application and **configures autoloading**. This directory also *houses* the **assets** such as **images**, **JavaScript**, and **CSS**.

Laravel Directory Structure

The App Directory

- # The `Broadcasting` Directory
- # The `Console` Directory
- # The `Events` Directory
- # The `Exceptions` Directory
- # The `Http` Directory
- # The `Jobs` Directory
- # The `Listeners` Directory
- # The `Mail` Directory
- # The `Models` Directory
- # The `Notifications` Directory
- # The `Policies` Directory
- # The `Providers` Directory
- # The `Rules` Directory

The Root Directory

- # The `app` Directory
- # The `bootstrap` Directory
- # The `config` Directory
- # The `database` Directory
- # The `public` Directory
- # The `resources` Directory
- # The `routes` Directory
- # The `storage` Directory
- # The `tests` Directory
- # The `vendor` Directory

Read Source:

<https://laravel.com/docs/8.x/structure/>

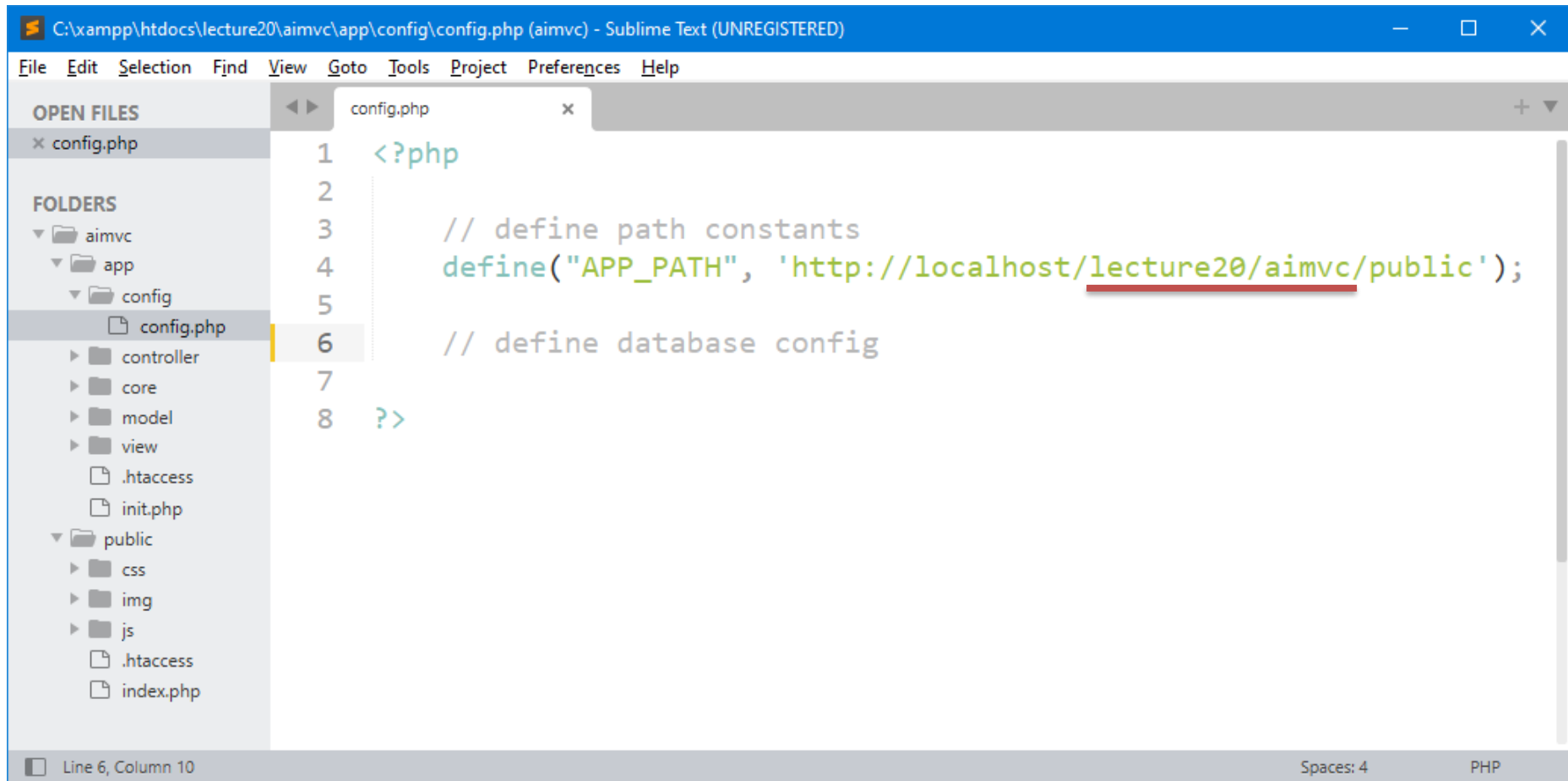


You're Started Here !!!

The slide features a white background with green leaves framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub, with some showing signs of aging or damage. The text is centered in the middle of the slide.

1 – Create URL Path Constants

1 – Create *config* File & Directory



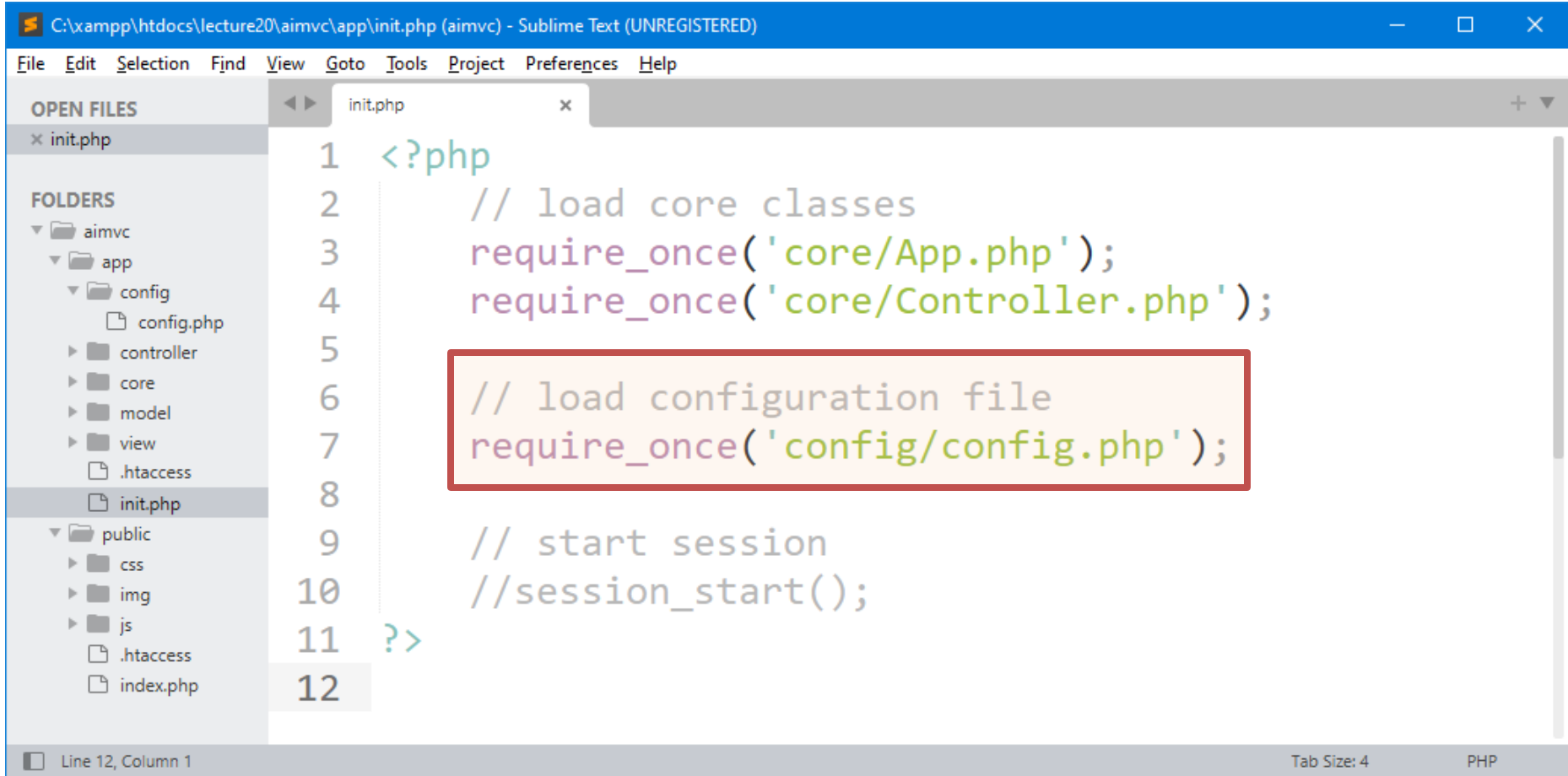
The screenshot shows the Sublime Text editor interface. The title bar indicates the file path is C:\xampp\htdocs\lecture20\aimvc\app\config\config.php (aimvc) - Sublime Text (UNREGISTERED). The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. On the left, the 'FOLDERS' sidebar shows a tree structure: aimvc > app > config > config.php. The main editor area displays the content of config.php with line numbers 1 through 8. The code is as follows:

```
1 <?php
2
3 // define path constants
4 define("APP_PATH", 'http://localhost/lecture20/aimvc/public');
5
6 // define database config
7
8 ?>
```

The status bar at the bottom shows 'Line 6, Column 10', 'Spaces: 4', and 'PHP'.

Define new *path constants* with the value is the your base URL.
Remember, if you change the **current working directory** name, you may need to rename also in this *config* file.

2 – Initialize *Config File* as a the Core



C:\xampp\htdocs\lecture20\aimvc\app\init.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x init.php

FOLDERS
▼ aimvc
▼ app
▼ config
config.php
▶ controller
▶ core
▶ model
▶ view
.htaccess
init.php
▼ public
css
img
js
.htaccess
index.php

```
1 <?php
2     // load core classes
3     require_once('core/App.php');
4     require_once('core/Controller.php');
5
6     // load configuration file
7     require_once('config/config.php');
8
9     // start session
10    //session_start();
11 ?>
12
```

Line 12, Column 1 Tab Size: 4 PHP

A decorative border of green leaves and branches at the top of the page.

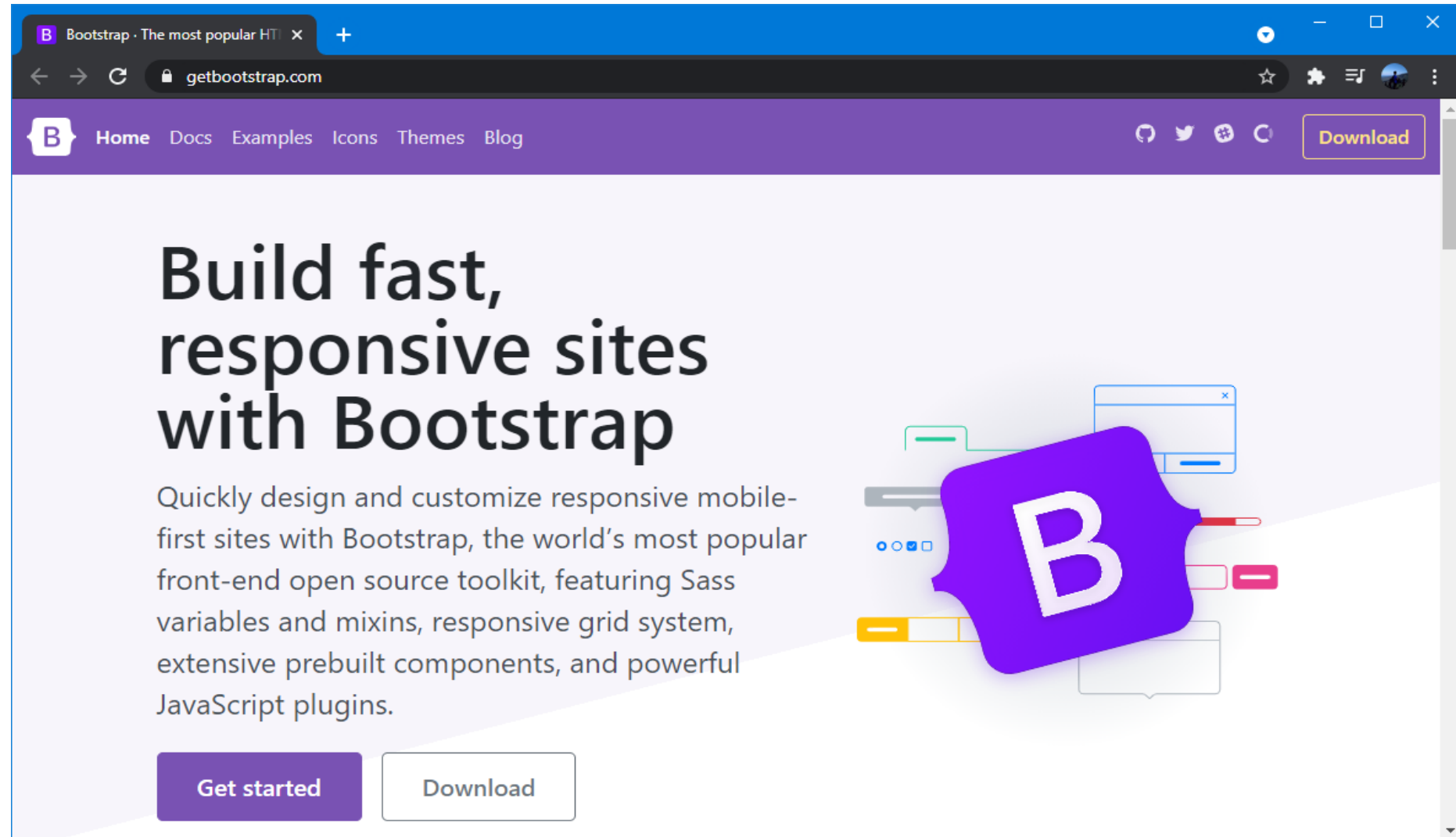
2 – Download *Bootstrap* and Put in Public/Assets Directory

A decorative border of green leaves and branches at the bottom of the page.

1 - Download Bootstrap

Visit Bootstrap from Bootstrap official website.

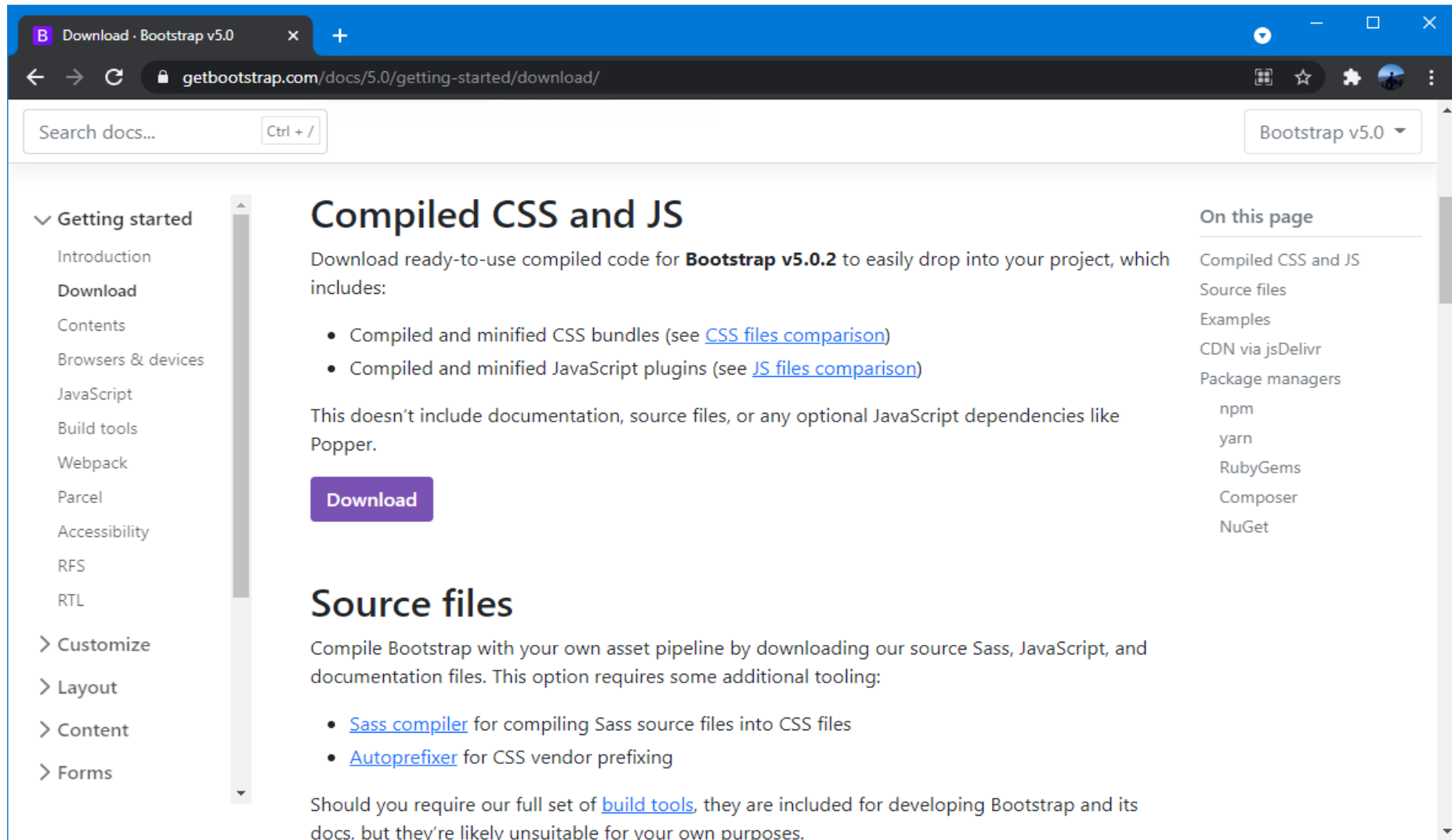
Link: <https://getbootstrap.com/>



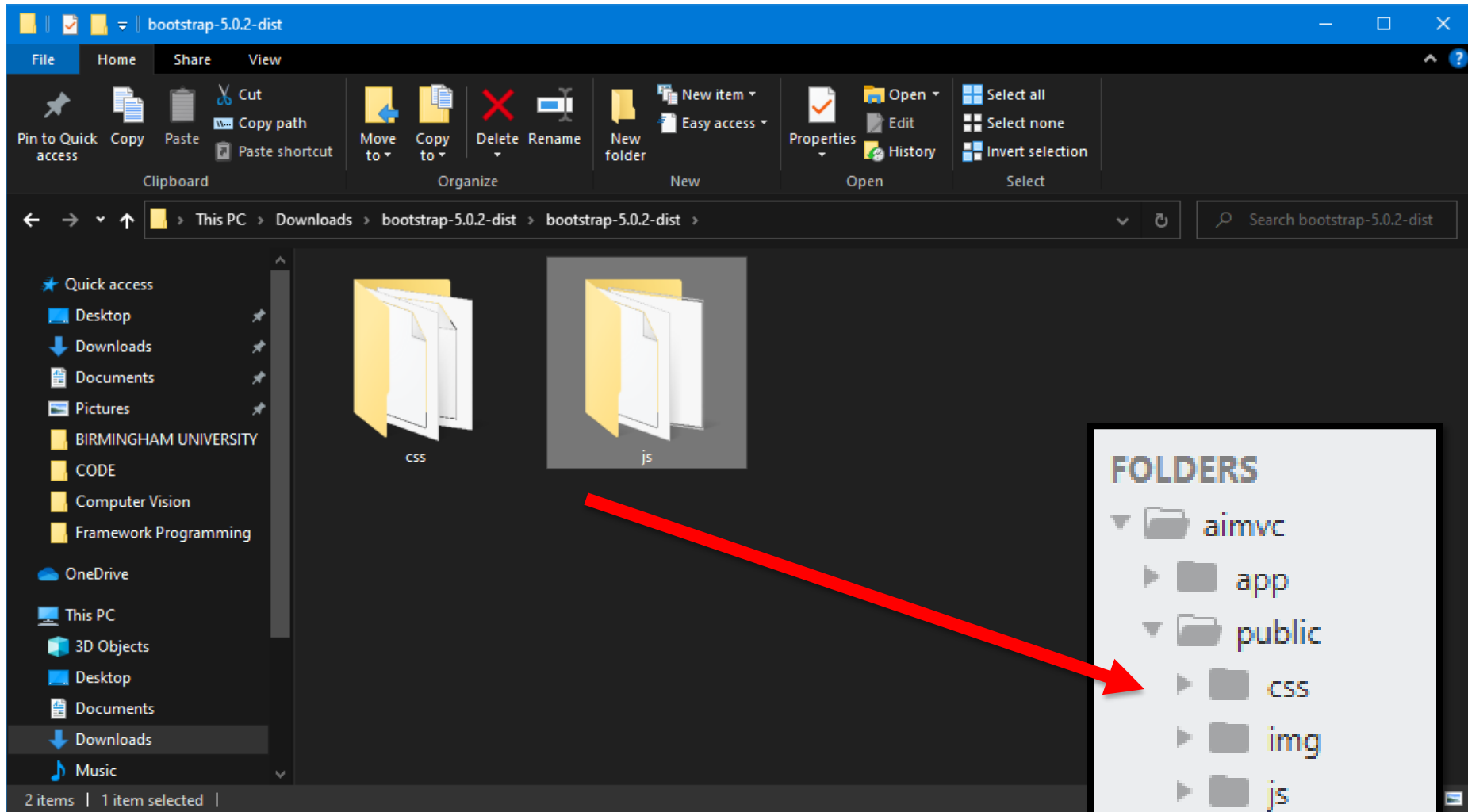
2 - Download Bootstrap

Download *ready-to-use compiled code* for **Bootstrap v5.0.2** to easily drop into your project.

Link: <https://getbootstrap.com/docs/5.0/getting-started/download/>

A screenshot of a web browser displaying the Bootstrap v5.0.2 download page. The browser's address bar shows the URL 'getbootstrap.com/docs/5.0/getting-started/download/'. The page has a blue header with the Bootstrap logo and version 'v5.0'. A search bar is located below the header. On the left, a sidebar menu lists various sections: 'Getting started' (with sub-items like Introduction, Download, Contents, etc.), 'Customize', 'Layout', 'Content', and 'Forms'. The main content area is titled 'Compiled CSS and JS' and contains text about downloading ready-to-use code for Bootstrap v5.0.2. It includes a bulleted list of what is included (compiled and minified CSS bundles and JavaScript plugins) and a 'Download' button. Below this, there is a section for 'Source files' which explains how to compile Bootstrap with your own asset pipeline. On the right side of the page, there is a 'On this page' section with links to 'Compiled CSS and JS', 'Source files', 'Examples', 'CDN via jsDelivr', and 'Package managers' (listing npm, yarn, RubyGems, Composer, and NuGet).

3 - Extract Bootstrap & Place *them* in Project



Copy and paste these two folders in our ***public/assets*** ***directory***, if the directory already existed, you can paste and replace both directories.

A decorative border of green leaves and branches at the top of the page.

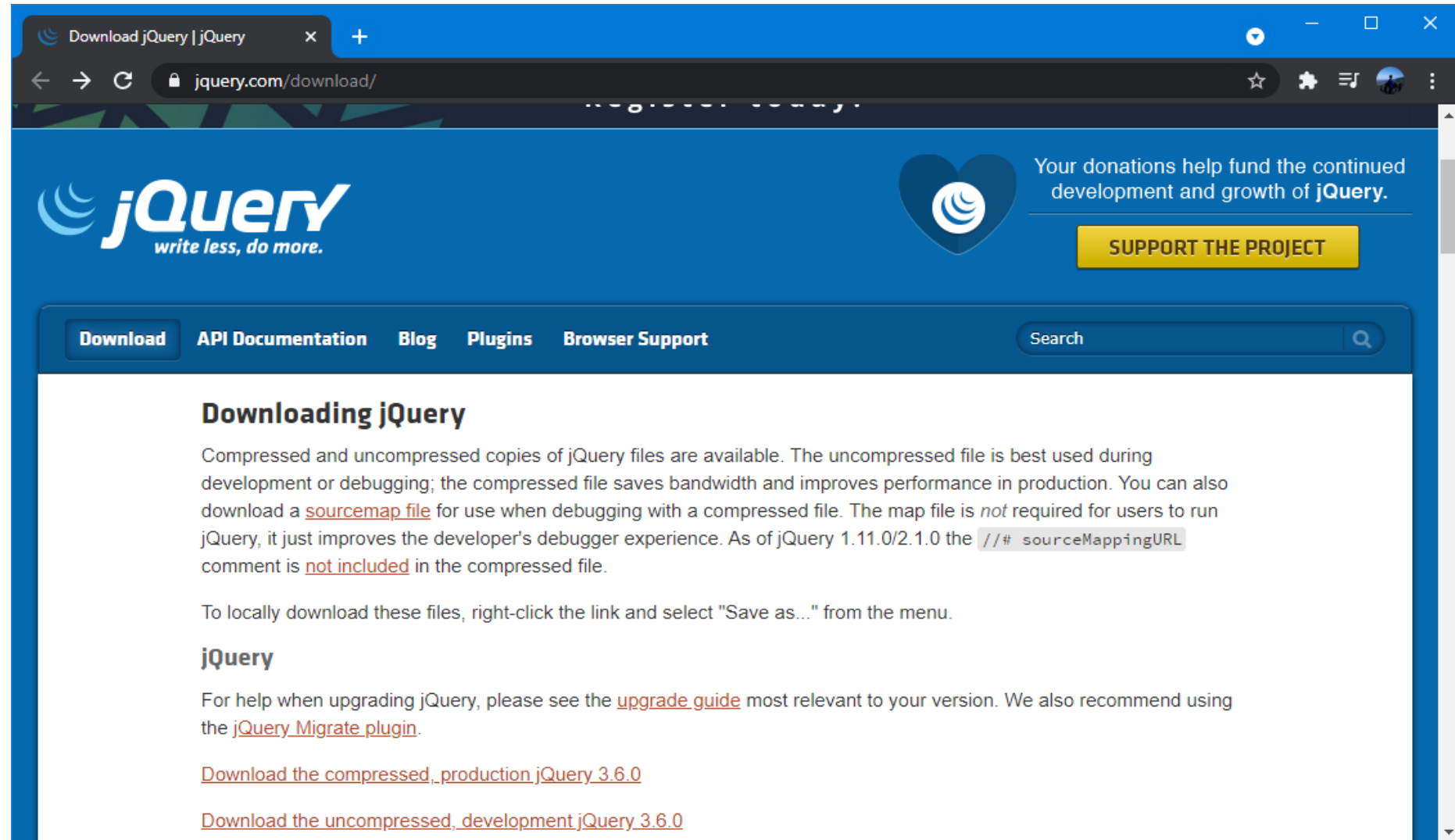
2 – Download *jQuery* and Put in Public/Assets Directory

A decorative border of green leaves and branches at the bottom of the page.

1 - Download jQuery

Visit jQuery from jQuery official website.

Link: <https://jquery.com/download/>

A screenshot of a web browser displaying the jQuery download page. The browser's address bar shows 'jquery.com/download/'. The page has a blue header with the jQuery logo and tagline 'write less, do more.' on the left. On the right, there is a heart icon and text stating 'Your donations help fund the continued development and growth of jQuery.' with a yellow 'SUPPORT THE PROJECT' button. Below the header is a navigation bar with links: 'Download', 'API Documentation', 'Blog', 'Plugins', and 'Browser Support'. A search bar is also present. The main content area is titled 'Downloading jQuery' and contains text explaining the availability of compressed and uncompressed files, the use of source maps, and a note about the sourceMappingURL comment. It also provides instructions on how to download the files locally. At the bottom, there are two links: 'Download the compressed, production jQuery 3.6.0' and 'Download the uncompressed, development jQuery 3.6.0'.

Download jQuery | jQuery

jquery.com/download/

jQuery
write less, do more.

Your donations help fund the continued development and growth of jQuery.

SUPPORT THE PROJECT

Download API Documentation Blog Plugins Browser Support

Search

Downloading jQuery

Compressed and uncompressed copies of jQuery files are available. The uncompressed file is best used during development or debugging; the compressed file saves bandwidth and improves performance in production. You can also download a [sourcemap file](#) for use when debugging with a compressed file. The map file is *not* required for users to run jQuery, it just improves the developer's debugger experience. As of jQuery 1.11.0/2.1.0 the `//# sourceMappingURL` comment is [not included](#) in the compressed file.

To locally download these files, right-click the link and select "Save as..." from the menu.

jQuery

For help when upgrading jQuery, please see the [upgrade guide](#) most relevant to your version. We also recommend using the [jQuery Migrate plugin](#).

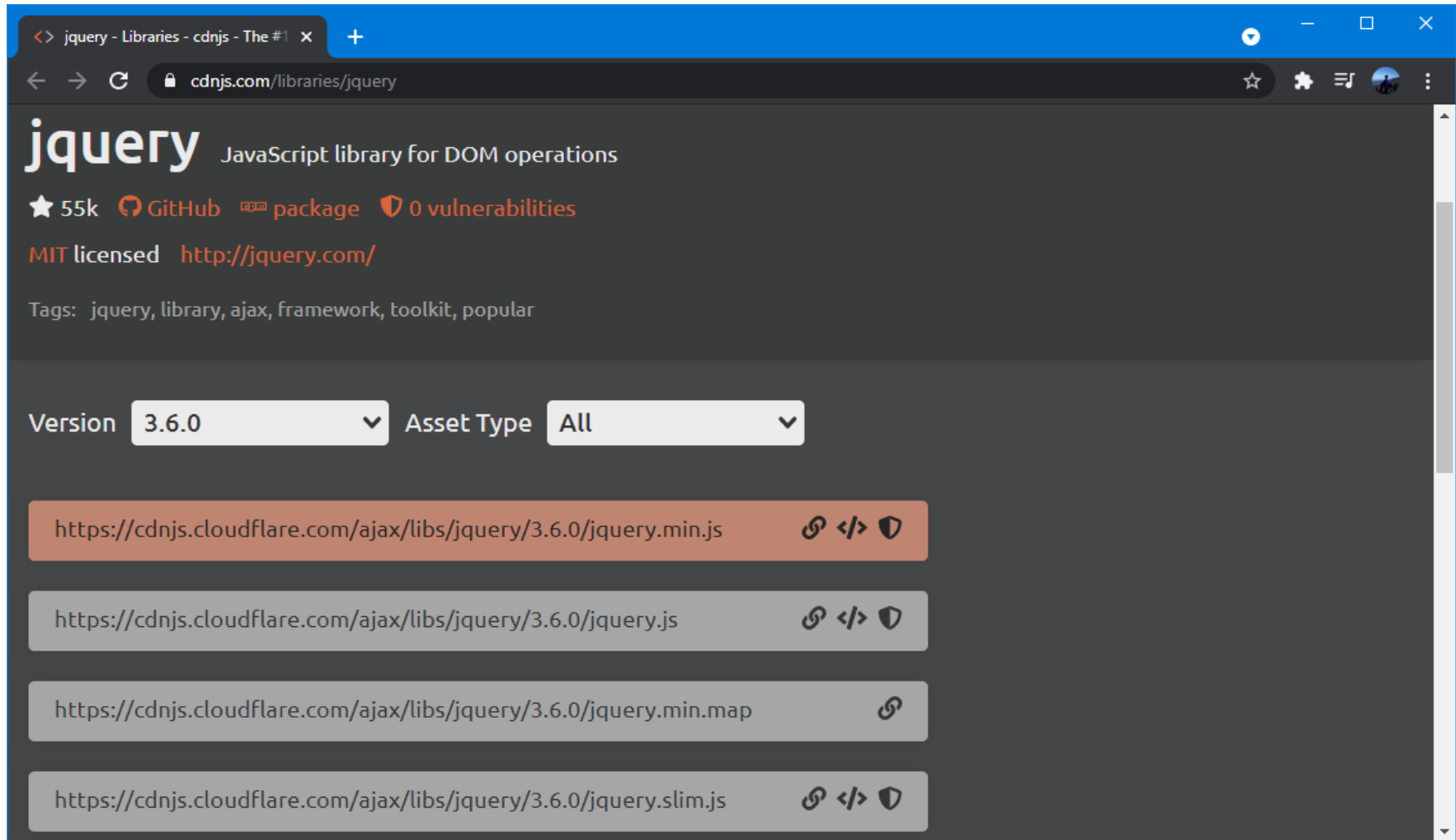
[Download the compressed, production jQuery 3.6.0](#)

[Download the uncompressed, development jQuery 3.6.0](#)

2 - Download *jQuery.slim.min.js*

Visit jQuery from jQuery official website.

Link: <https://cdnjs.com/libraries/jquery/>
<https://code.jquery.com/jquery-3.3.1.slim.min.js>



3 – Save to *your* Working Directory

The screenshot shows a Windows File Explorer window with the address bar displaying the path: `This PC > OS (C:) > xampp > htdocs > lecture20 > aimvc > public > js`. The left sidebar shows the 'Quick access' pane with various locations like Desktop, Downloads, Documents, Pictures, and OneDrive. The main pane displays a list of files with columns for Name, Date modified, Type, and Size. The file `jquery-3.3.1.slim.min.js` is selected and highlighted.

Name	Date modified	Type	Size
bootstrap.bundle.js	6/23/2021 2:25 AM	JavaScript File	204 KB
bootstrap.bundle.js.map	6/23/2021 2:25 AM	MAP File	415 KB
bootstrap.bundle.min.js	6/23/2021 2:25 AM	JavaScript File	77 KB
bootstrap.bundle.min.js.map	6/23/2021 2:25 AM	MAP File	319 KB
bootstrap.esm.js	6/23/2021 2:25 AM	JavaScript File	137 KB
bootstrap.esm.js.map	6/23/2021 2:25 AM	MAP File	283 KB
bootstrap.esm.min.js	6/23/2021 2:25 AM	JavaScript File	72 KB
bootstrap.esm.min.js.map	6/23/2021 2:25 AM	MAP File	218 KB
bootstrap.js	6/23/2021 2:25 AM	JavaScript File	146 KB
bootstrap.js.map	6/23/2021 2:25 AM	MAP File	284 KB
bootstrap.min.js	6/23/2021 2:25 AM	JavaScript File	59 KB
bootstrap.min.js.map	6/23/2021 2:25 AM	MAP File	212 KB
jquery-3.3.1.slim.min.js	6/24/2021 11:52 AM	JavaScript File	69 KB

13 items | 1 item selected 68.2 KB

Does bootstrap 5 require jQuery?

Bootstrap 5 is designed to be used ***without*** jQuery, but it's still possible to use our components with jQuery.

Bootstrap uses jQuery for JavaScript plugins (***like modals, tooltips, etc***). However, if you just use the CSS part of Bootstrap, you don't need jQuery.

A decorative border of green leaves and branches at the top of the slide.

3 – Quick start to add *Bootstrap* & *jQuery* to Back-end Framework

A decorative border of green leaves and branches at the bottom of the slide.

1 - Get started with Bootstrap

The screenshot shows the Bootstrap v5.0 documentation page. The browser's address bar displays the URL `getbootstrap.com/docs/5.0/getting-started/introduction/`. The page has a dark blue header with the Bootstrap logo and version information. A search bar is located at the top left, and a dropdown menu for 'Bootstrap v5.0' is at the top right. The main content area is divided into three sections: 'CSS', 'JS', and 'Bundle'. Each section contains a code block with a 'Copy' button. The 'CSS' section shows the `<link href='\"https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css\"'>` code. The 'JS' section shows the `<script src='\"https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.bundle.min.js\"'>` code. The 'Bundle' section shows the `<script src='\"https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.bundle.min.js\"'>` code. On the right side, there is a 'On this page' sidebar with links to 'Quick start', 'CSS', 'JS', 'Bundle', 'Separate', 'Modules', 'Components', 'Starter template', 'Important globals', 'HTML5 doctype', 'Responsive meta tag', 'Box-sizing', 'Reboot', and 'Community'. The left sidebar contains a 'Getting started' section with links to 'Introduction', 'Download', 'Contents', 'Browsers & devices', 'JavaScript', 'Build tools', 'Webpack', 'Parcel', 'Accessibility', 'RFS', 'RTL', 'Customize', 'Layout', 'Content', and 'Forms'.

Introduction · Bootstrap v5.0

getbootstrap.com/docs/5.0/getting-started/introduction/

Search docs... Ctrl + /

Bootstrap v5.0

CSS

Copy-paste the stylesheet `<link>` into your `<head>` before all other stylesheets to load our CSS.

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css">
```

Copy

JS

Many of our components require the use of JavaScript to function. Specifically, they require our own JavaScript plugins and [Popper](#). Place **one of the following** `<script>s` near the end of your pages, right before the closing `</body>` tag, to enable them.

Bundle

Include every Bootstrap JavaScript plugin and dependency with one of our two bundles. Both `bootstrap.bundle.js` and `bootstrap.bundle.min.js` include [Popper](#) for our tooltips and popovers. For more information about what's included in Bootstrap, please see our [contents](#) section.

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.bundle.min.js">
```

Copy

On this page

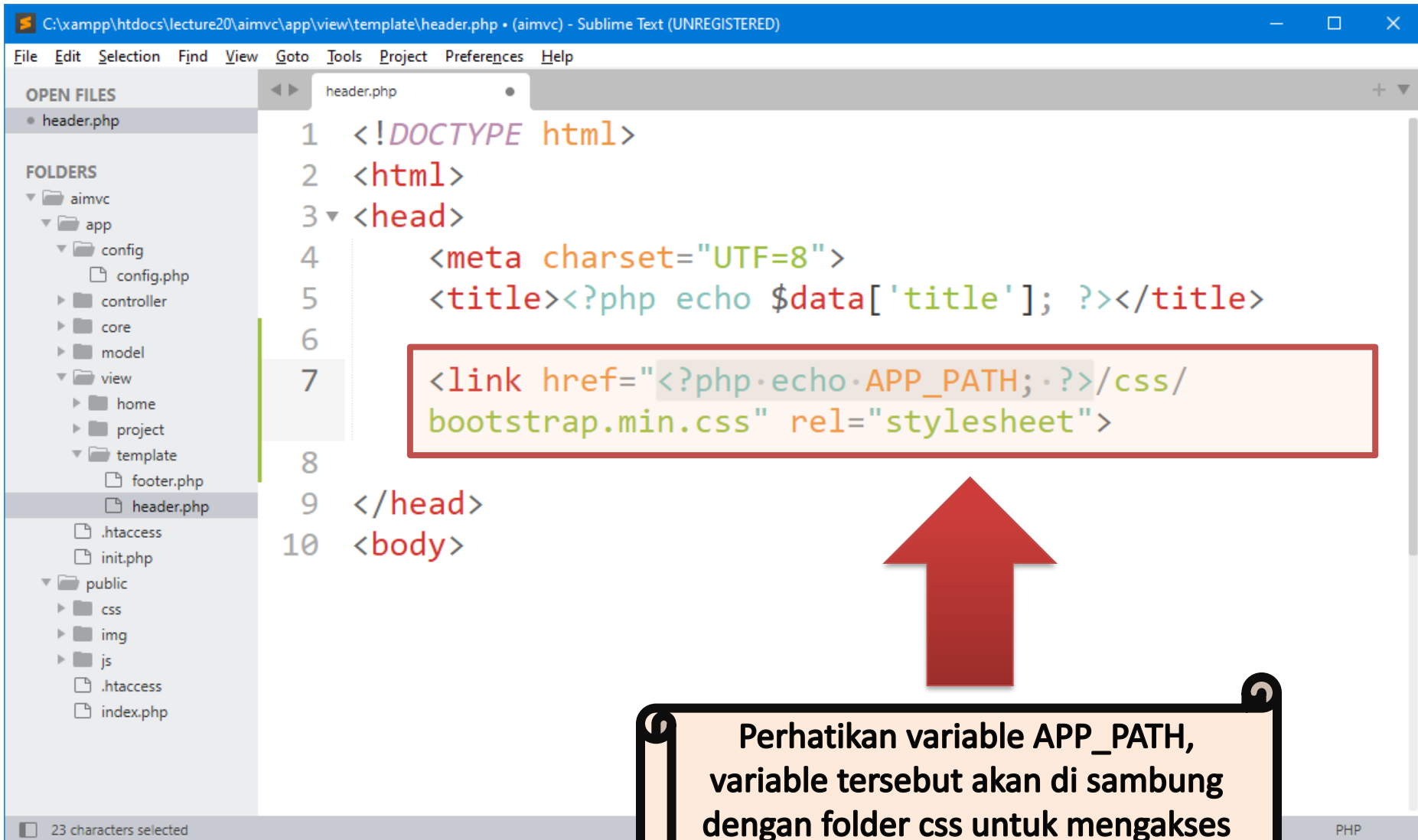
- Quick start
 - CSS
 - JS
 - Bundle
 - Separate
 - Modules
 - Components
- Starter template
- Important globals
 - HTML5 doctype
 - Responsive meta tag
 - Box-sizing
 - Reboot
- Community

Getting started

- Introduction
- Download
- Contents
- Browsers & devices
- JavaScript
- Build tools
- Webpack
- Parcel
- Accessibility
- RFS
- RTL
- Customize
- Layout
- Content
- Forms

Link: <https://getbootstrap.com/docs/5.0/getting-started/introduction/>

2 – Place *Bootstrap* in Project (*Offline*)



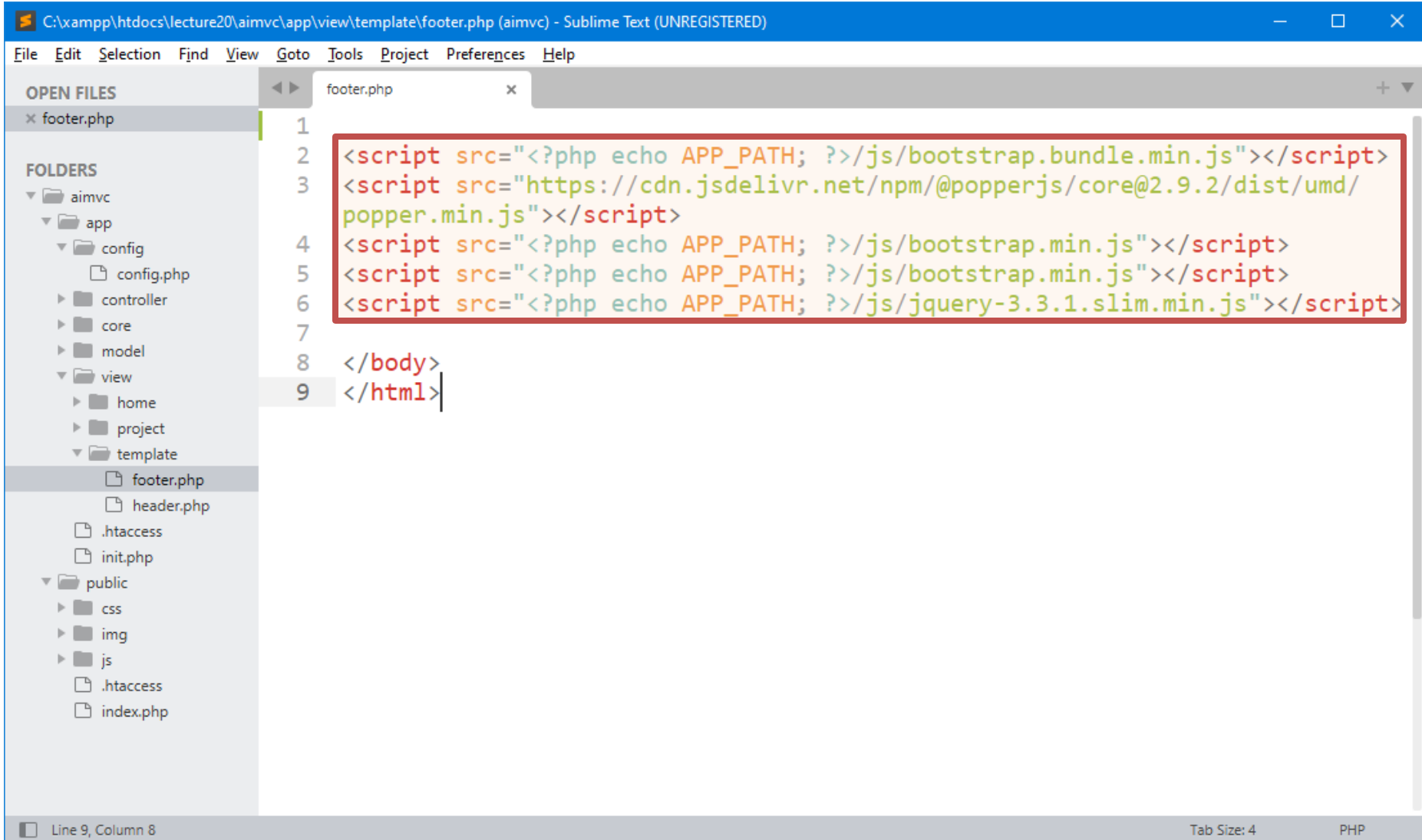
The screenshot shows a Sublime Text editor window with the file path `C:\xampp\htdocs\lecture20\aimvc\app\view\template\header.php`. The editor displays the following PHP code in `header.php`:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title><?php echo $data['title']; ?></title>
6
7     <link href="<?php·echo·APP_PATH;·?>/css/
8       bootstrap.min.css" rel="stylesheet">
9
10 </head>
11 <body>
```

The code on line 7 is highlighted with a red box. A large red arrow points from a callout box below to this line.

Perhatikan variable APP_PATH, variable tersebut akan di sambung dengan folder css untuk mengakses file bootstrap.

3 – Place *JavaScript* in Project

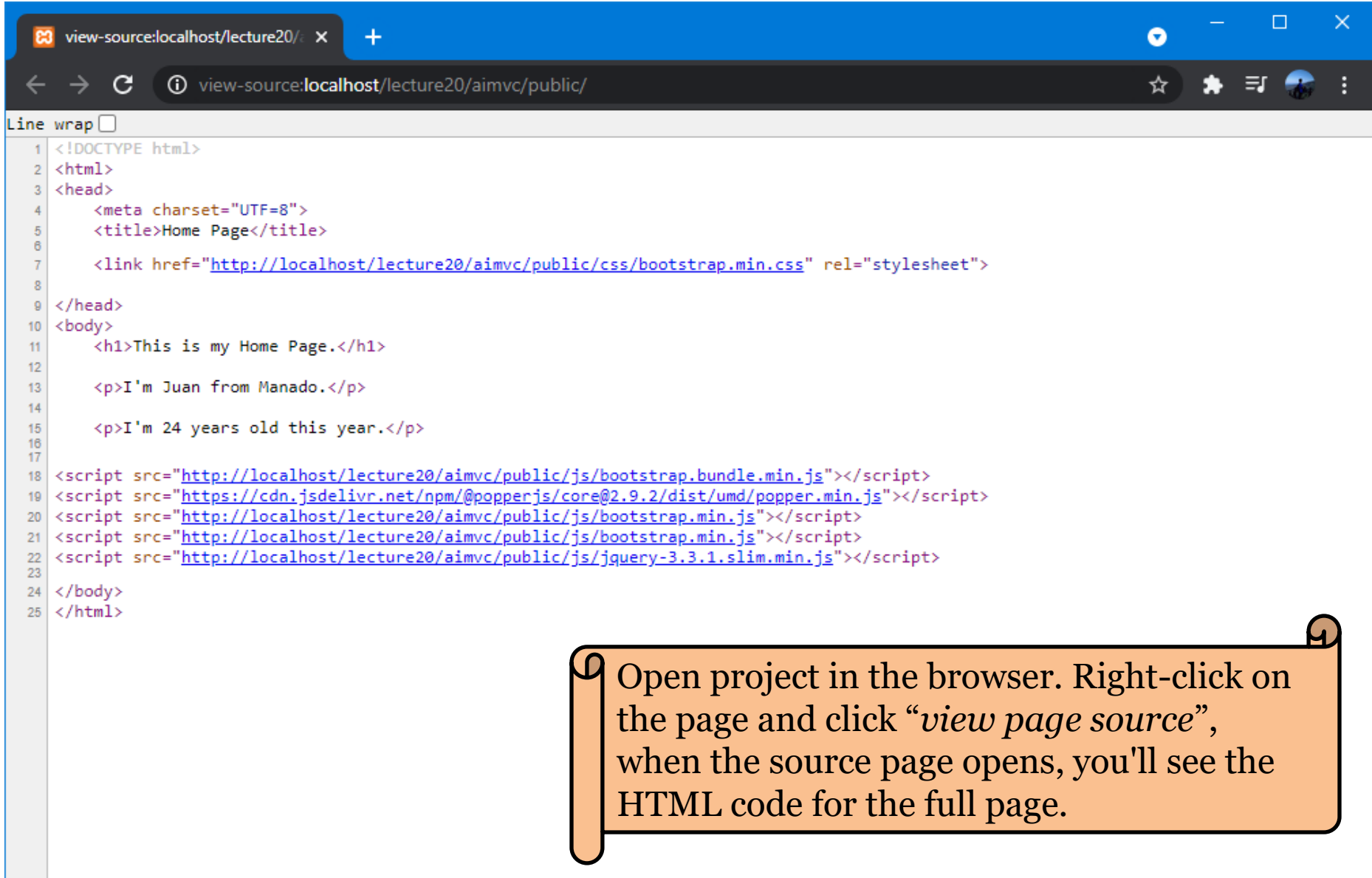


```
C:\xampp\htdocs\lecture20\aimvc\app\view\template\footer.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
OPEN FILES
x footer.php
FOLDERS
aimvc
  app
    config
      config.php
    controller
    core
    model
    view
      home
      project
      template
        footer.php
        header.php
      .htaccess
      init.php
    public
      css
      img
      js
      .htaccess
      index.php
  footer.php
  header.php
  .htaccess
  init.php
  public
    css
    img
    js
    .htaccess
    index.php

1
2 <script src="<?php echo APP_PATH; ?>/js/bootstrap.bundle.min.js"></script>
3 <script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.9.2/dist/umd/
  popper.min.js"></script>
4 <script src="<?php echo APP_PATH; ?>/js/bootstrap.min.js"></script>
5 <script src="<?php echo APP_PATH; ?>/js/bootstrap.min.js"></script>
6 <script src="<?php echo APP_PATH; ?>/js/jquery-3.3.1.slim.min.js"></script>
7
8 </body>
9 </html>
```

Line 9, Column 8 Tab Size: 4 PHP

4 – Testing Bootstrap



view-source:localhost/lecture20/ X +

view-source:localhost/lecture20/aimvc/public/ ☆ ⚙️ 🎵 👤 ⋮

Line wrap ☐

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="UTF-8">
5   <title>Home Page</title>
6
7   <link href="http://localhost/lecture20/aimvc/public/css/bootstrap.min.css" rel="stylesheet">
8
9 </head>
10 <body>
11   <h1>This is my Home Page.</h1>
12
13   <p>I'm Juan from Manado.</p>
14
15   <p>I'm 24 years old this year.</p>
16
17
18 <script src="http://localhost/lecture20/aimvc/public/js/bootstrap.bundle.min.js"></script>
19 <script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.9.2/dist/umd/popper.min.js"></script>
20 <script src="http://localhost/lecture20/aimvc/public/js/bootstrap.min.js"></script>
21 <script src="http://localhost/lecture20/aimvc/public/js/bootstrap.min.js"></script>
22 <script src="http://localhost/lecture20/aimvc/public/js/jquery-3.3.1.slim.min.js"></script>
23
24 </body>
25 </html>
```

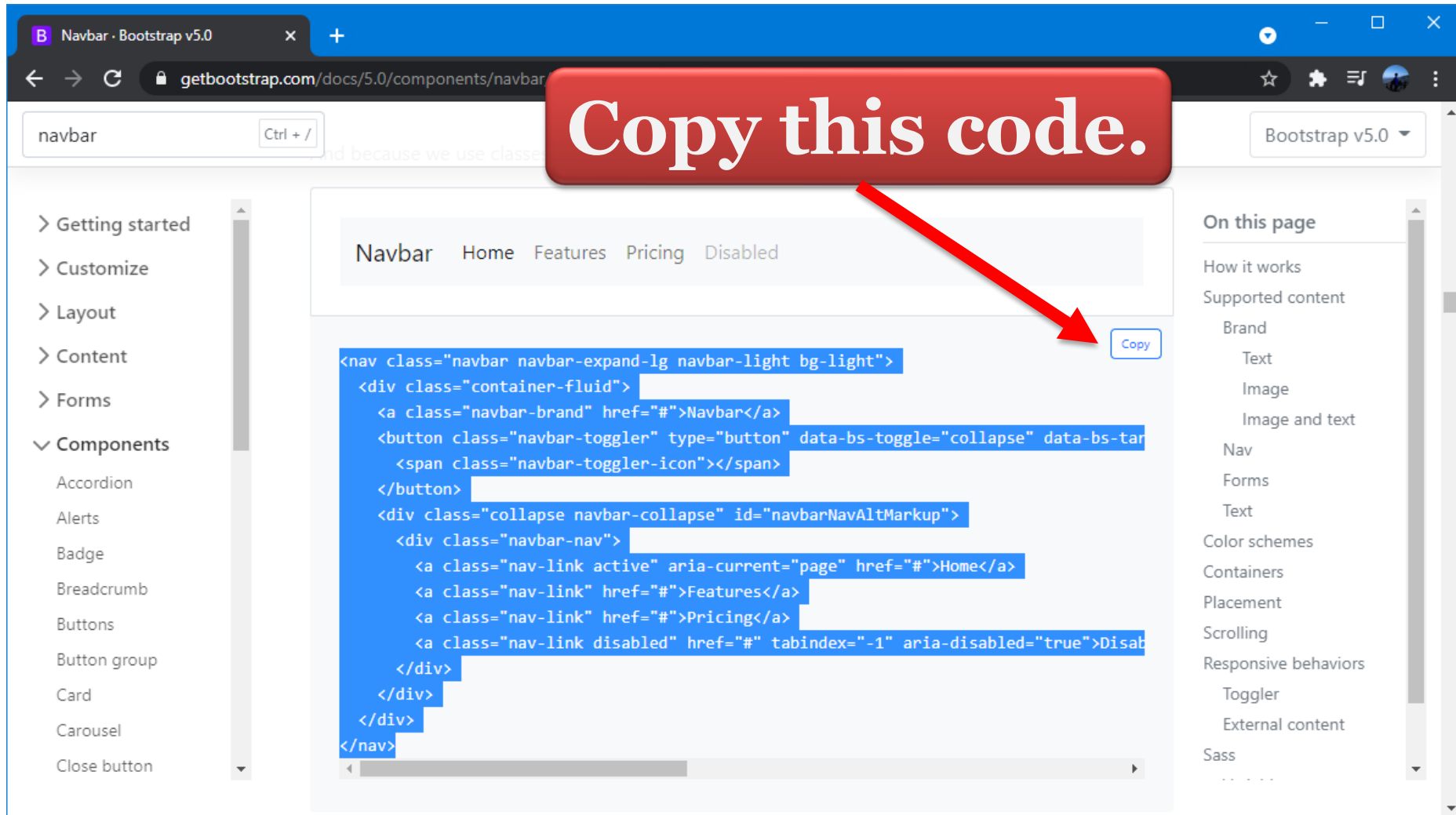
Open project in the browser. Right-click on the page and click “*view page source*”, when the source page opens, you'll see the HTML code for the full page.

A decorative border of green leaves and branches at the top of the page.

4 – Create *a* Simple Bootstrap Page with navigation header (*Navbar*)

A decorative border of green leaves and branches at the bottom of the page.

1 – Find Example *Navbar* in Bootstrap



The image shows a web browser window displaying the Bootstrap v5.0 documentation page for the Navbar component. The browser's address bar shows the URL `getbootstrap.com/docs/5.0/components/navbar/`. A red callout box with the text "Copy this code." is overlaid on the page, with a red arrow pointing to a "Copy" button next to the code block. The code block contains the following HTML:

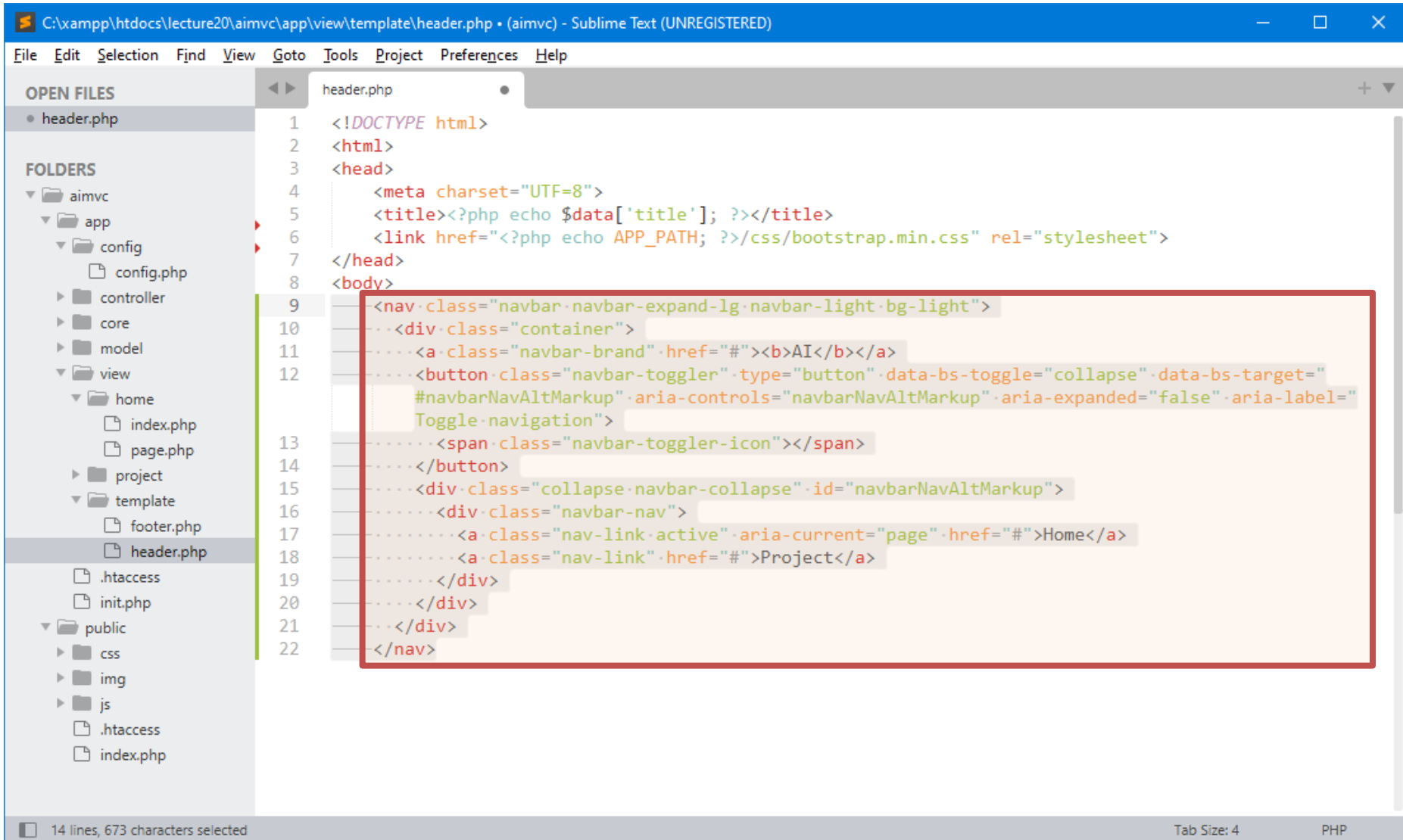
```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">Navbar</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav" data-bs-permitting="true">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <div class="navbar-nav">
        <a class="nav-link active" aria-current="page" href="#">Home</a>
        <a class="nav-link" href="#">Features</a>
        <a class="nav-link" href="#">Pricing</a>
        <a class="nav-link disabled" href="#" tabindex="-1" aria-disabled="true">Disabled</a>
      </div>
    </div>
  </div>
</nav>
```

The page also features a sidebar on the left with a "Components" section expanded, and a "On this page" sidebar on the right listing various topics like "How it works", "Supported content", "Brand", "Text", "Image", "Image and text", "Nav", "Forms", "Text", "Color schemes", "Containers", "Placement", "Scrolling", "Responsive behaviors", "Toggler", "External content", and "Sass".

In column search, you can find Navbar or follow this link:

Link: <https://getbootstrap.com/docs/5.0/components/navbar/>

2 – Add Navbar in *header.php*



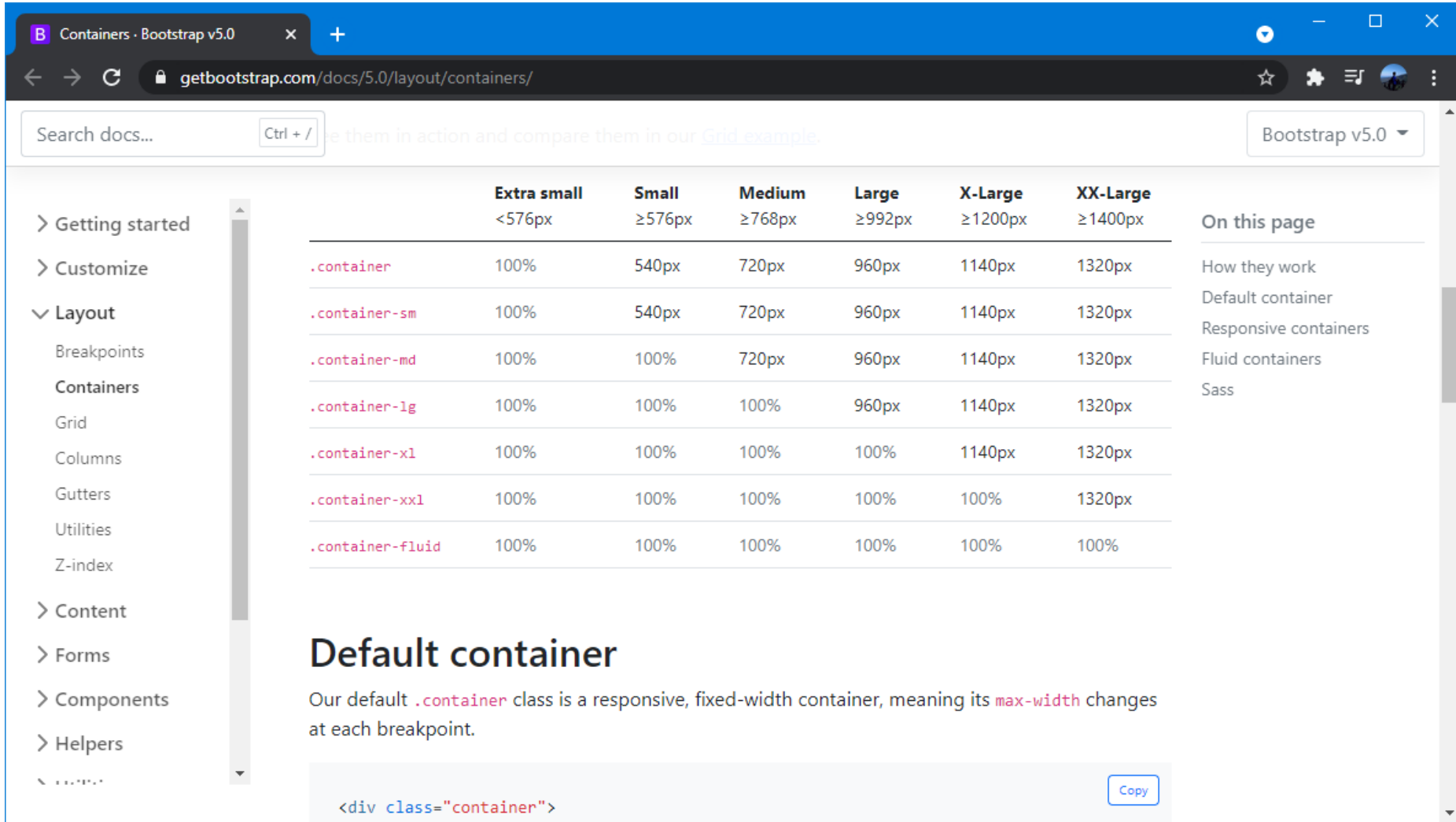
The screenshot shows the Sublime Text editor with the file `header.php` open. The editor's interface includes a menu bar (File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, Help), a sidebar with 'OPEN FILES' and 'FOLDERS' panels, and a main text area. The 'FOLDERS' panel shows a project structure with folders like `aimvc`, `app`, `config`, `controller`, `core`, `model`, `view`, `home`, `project`, `template`, `public`, `css`, `img`, and `js`. The `header.php` file is selected in the 'OPEN FILES' panel.

The code in `header.php` is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title><?php echo $data['title']; ?></title>
6     <link href="<?php echo APP_PATH; ?>/css/bootstrap.min.css" rel="stylesheet">
7 </head>
8 <body>
9     <nav class="navbar navbar-expand-lg navbar-light bg-light">
10         <div class="container">
11             <a class="navbar-brand" href="#"><b>AI</b></a>
12             <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="
13                 #navbarNavAltMarkup" aria-controls="navbarNavAltMarkup" aria-expanded="false" aria-label="
14                 Toggle navigation">
15                 <span class="navbar-toggler-icon"></span>
16             </button>
17             <div class="collapse navbar-collapse" id="navbarNavAltMarkup">
18                 <div class="navbar-nav">
19                     <a class="nav-link active" aria-current="page" href="#">Home</a>
20                     <a class="nav-link" href="#">Project</a>
21                 </div>
22             </div>
23         </div>
24     </nav>
```

The navbar code (lines 9-24) is highlighted with a red box. The status bar at the bottom indicates '14 lines, 673 characters selected' and 'Tab Size: 4 PHP'.

Please Check type of Bootstrap Containers



The screenshot shows the Bootstrap v5.0 documentation page for containers. The browser address bar shows the URL <https://getbootstrap.com/docs/5.0/layout/containers/>. The page features a search bar, a sidebar with navigation links, a table of container classes and their widths, and a section for the default container.

Search docs... Ctrl + / Bootstrap v5.0

On this page

- How they work
- Default container
- Responsive containers
- Fluid containers
- Sass

	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	X-Large ≥1200px	XX-Large ≥1400px
<code>.container</code>	100%	540px	720px	960px	1140px	1320px
<code>.container-sm</code>	100%	540px	720px	960px	1140px	1320px
<code>.container-md</code>	100%	100%	720px	960px	1140px	1320px
<code>.container-lg</code>	100%	100%	100%	960px	1140px	1320px
<code>.container-xl</code>	100%	100%	100%	100%	1140px	1320px
<code>.container-xxl</code>	100%	100%	100%	100%	100%	1320px
<code>.container-fluid</code>	100%	100%	100%	100%	100%	100%

Default container

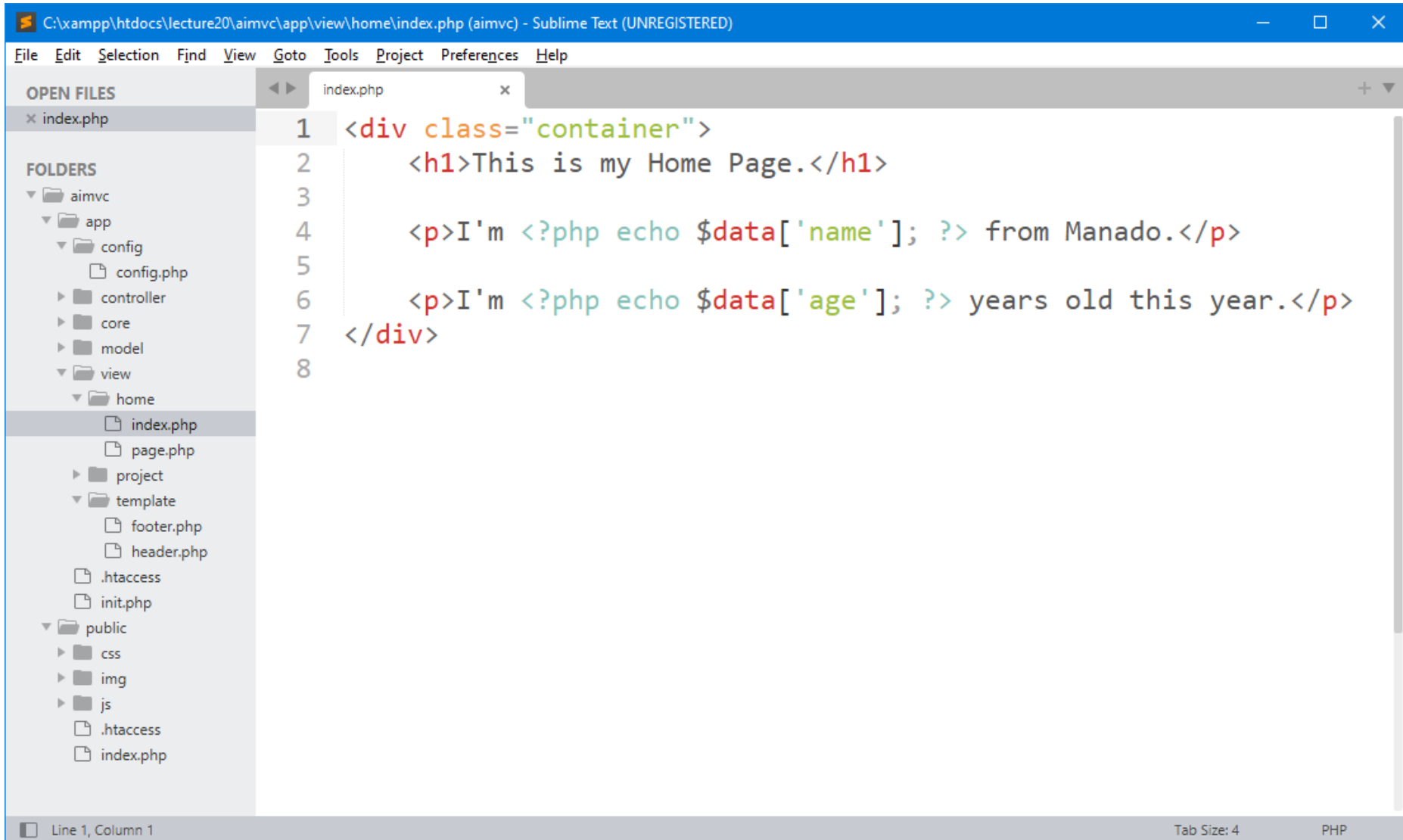
Our default `.container` class is a responsive, fixed-width container, meaning its `max-width` changes at each breakpoint.

```
<div class="container">
```

Copy

Link: <https://getbootstrap.com/docs/5.0/layout/containers/>

3 – Add Container in */view/home/index.php*



```
C:\xampp\htdocs\lecture20\aimvc\app\view\home\index.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x index.php

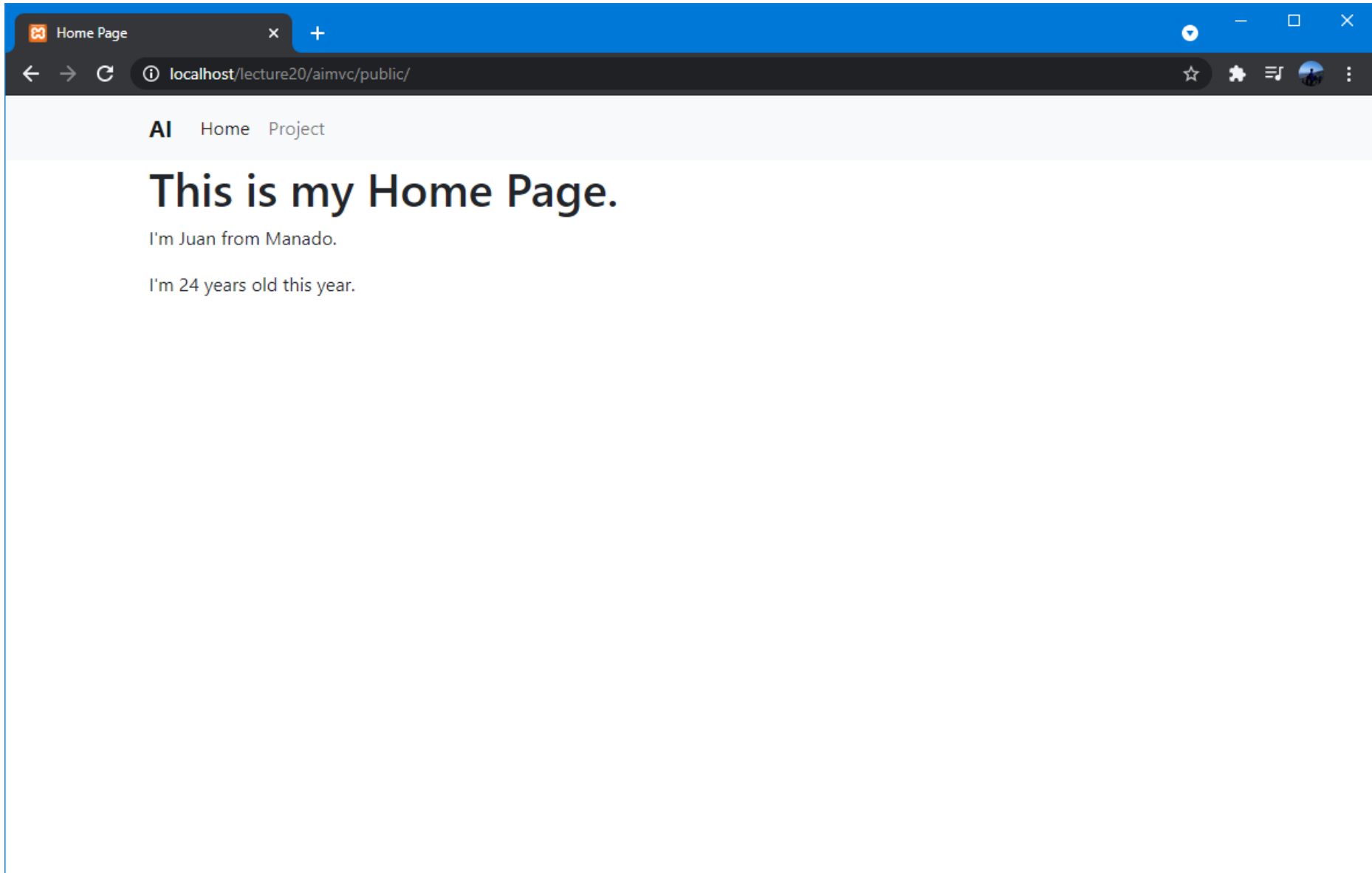
FOLDERS
aimvc
├── app
│   ├── config
│   │   └── config.php
│   ├── controller
│   ├── core
│   ├── model
│   └── view
│       └── home
│           ├── index.php
│           └── page.php
├── project
└── template
    ├── footer.php
    ├── header.php
    ├── .htaccess
    └── init.php
public
├── css
├── img
├── js
├── .htaccess
└── index.php

1 <div class="container">
2     <h1>This is my Home Page.</h1>
3
4     <p>I'm <?php echo $data['name']; ?> from Manado.</p>
5
6     <p>I'm <?php echo $data['age']; ?> years old this year.</p>
7 </div>
8

Line 1, Column 1
Tab Size: 4
PHP
```

The **<div>** tag defines a *division* or a section in an HTML document.

4 – Testing Bootstrap



A decorative border of green leaves and branches at the top of the slide.

5 – Create *link* in Navigation header (*Navbar*)

A decorative border of green leaves and branches at the bottom of the slide.

1 – All Links in Navbar

C:\xampp\htdocs\lecture20\aimvc\app\view\template\header.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x header.php

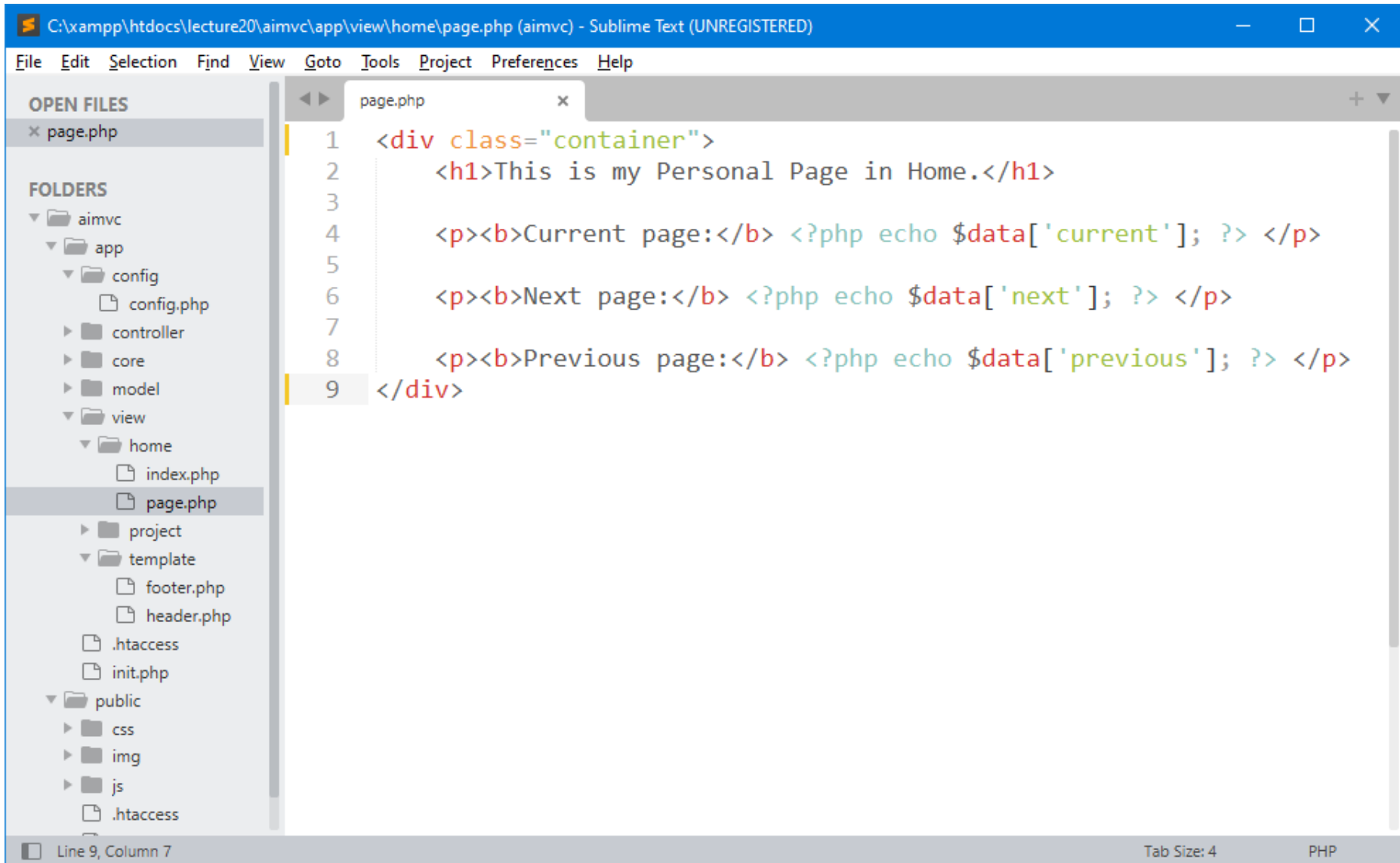
FOLDERS
aimvc
app
config
controller
core
model
view
home
index.php
page.php
project
template
footer.php
header.php
.htaccess
init.php
public
css
img
js
.htaccess
index.php

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF-8">
5     <title><?php echo $data['title']; ?></title>
6     <link href="<?php echo APP_PATH; ?>/css/bootstrap.min.css" rel="stylesheet">
7 </head>
8 <body>
9     <nav class="navbar navbar-expand-lg navbar-light bg-light">
10         <div class="container">
11             <a class="navbar-brand" href="<?php echo APP_PATH; ?>"><b>AI</b></a>
12             <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="
#navbarNavAltMarkup" aria-controls="navbarNavAltMarkup" aria-expanded="false" aria-label="
Toggle navigation">
13                 <span class="navbar-toggler-icon"></span>
14             </button>
15             <div class="collapse navbar-collapse" id="navbarNavAltMarkup">
16                 <div class="navbar-nav">
17                     <a class="nav-link active" aria-current="page" href="<?php echo APP_PATH; ?>">Home</a>
18                     <a class="nav-link" href="<?php echo APP_PATH; ?>/home/page/">Page</a>
19                     <a class="nav-link" href="<?php echo APP_PATH; ?>/project/">Project</a>
20                 </div>
21             </div>
22         </div>
23     </nav>
```

4 selection regions

Tab Size: 4 PHP

2 – Don't forget add *container* in *page.php* file



```
C:\xampp\htdocs\lecture20\aimvc\app\view\home\page.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

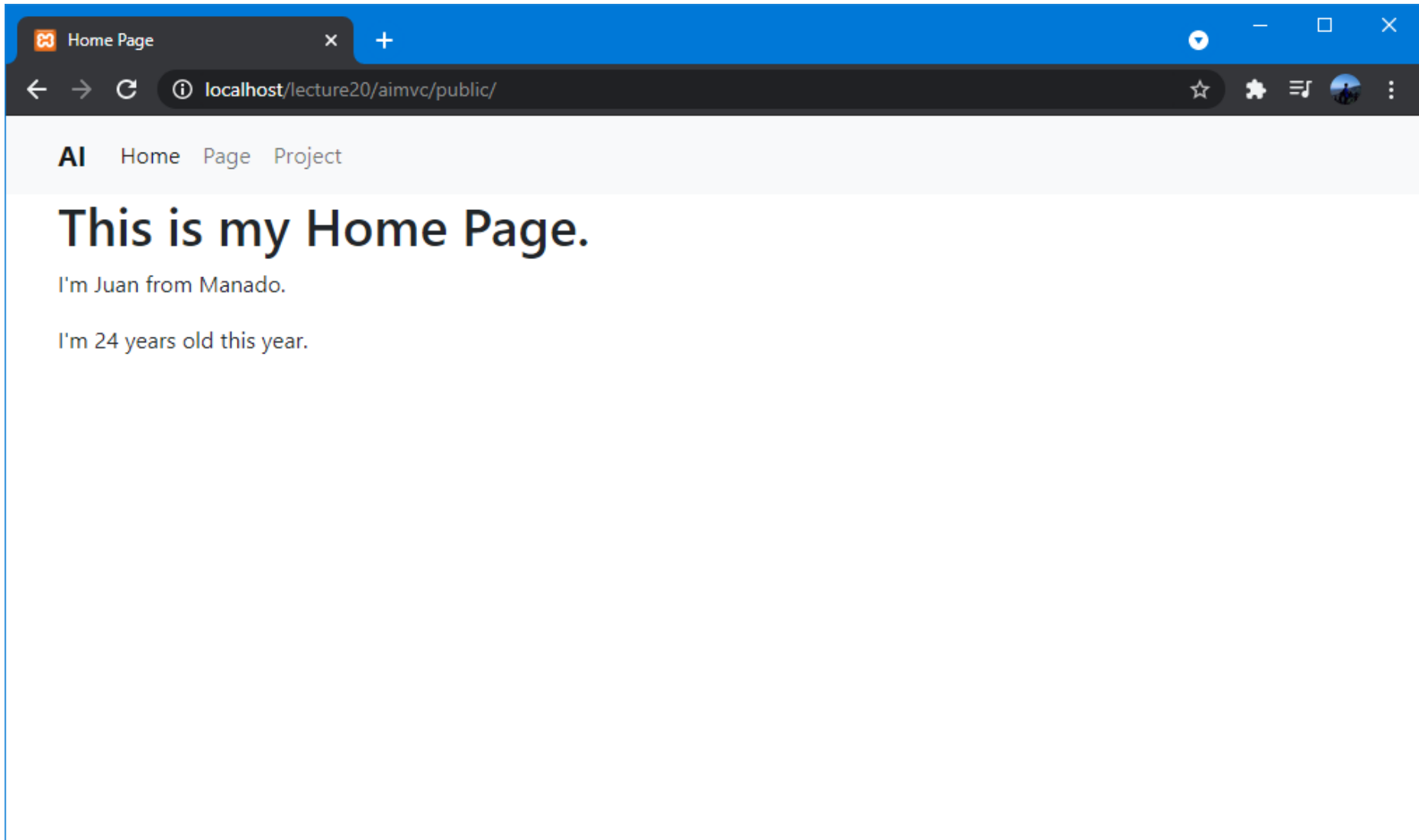
OPEN FILES
x page.php

FOLDERS
aimvc
├── app
│   ├── config
│   │   └── config.php
│   ├── controller
│   ├── core
│   ├── model
│   └── view
│       ├── home
│       │   ├── index.php
│       │   └── page.php
│       ├── project
│       └── template
│           ├── footer.php
│           └── header.php
├── .htaccess
├── init.php
├── public
│   ├── css
│   ├── img
│   ├── js
│   └── .htaccess
└── ...

1 <div class="container">
2     <h1>This is my Personal Page in Home.</h1>
3
4     <p><b>Current page:</b> <?php echo $data['current']; ?> </p>
5
6     <p><b>Next page:</b> <?php echo $data['next']; ?> </p>
7
8     <p><b>Previous page:</b> <?php echo $data['previous']; ?> </p>
9 </div>
```

Line 9, Column 7 Tab Size: 4 PHP

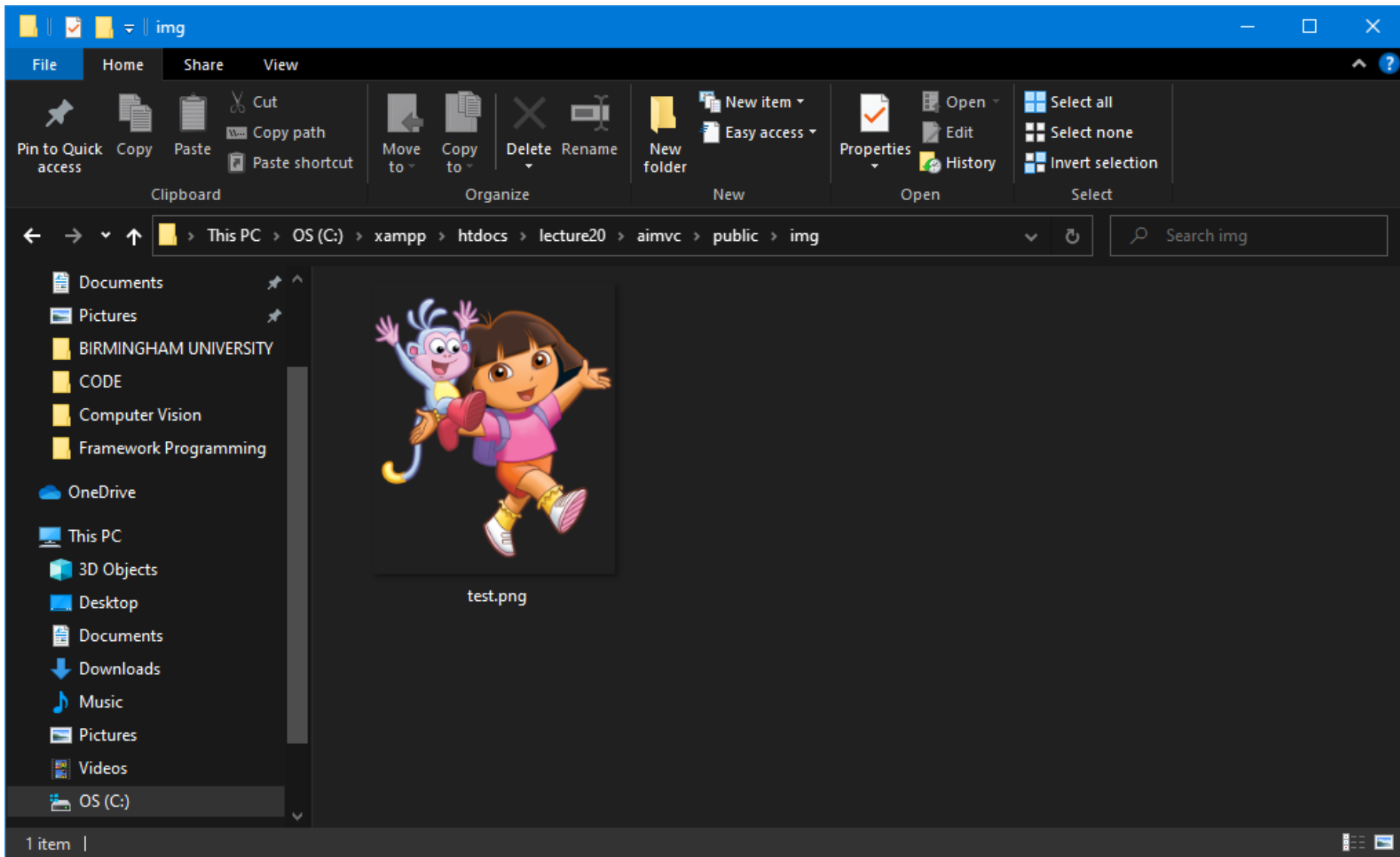
3 – Testing



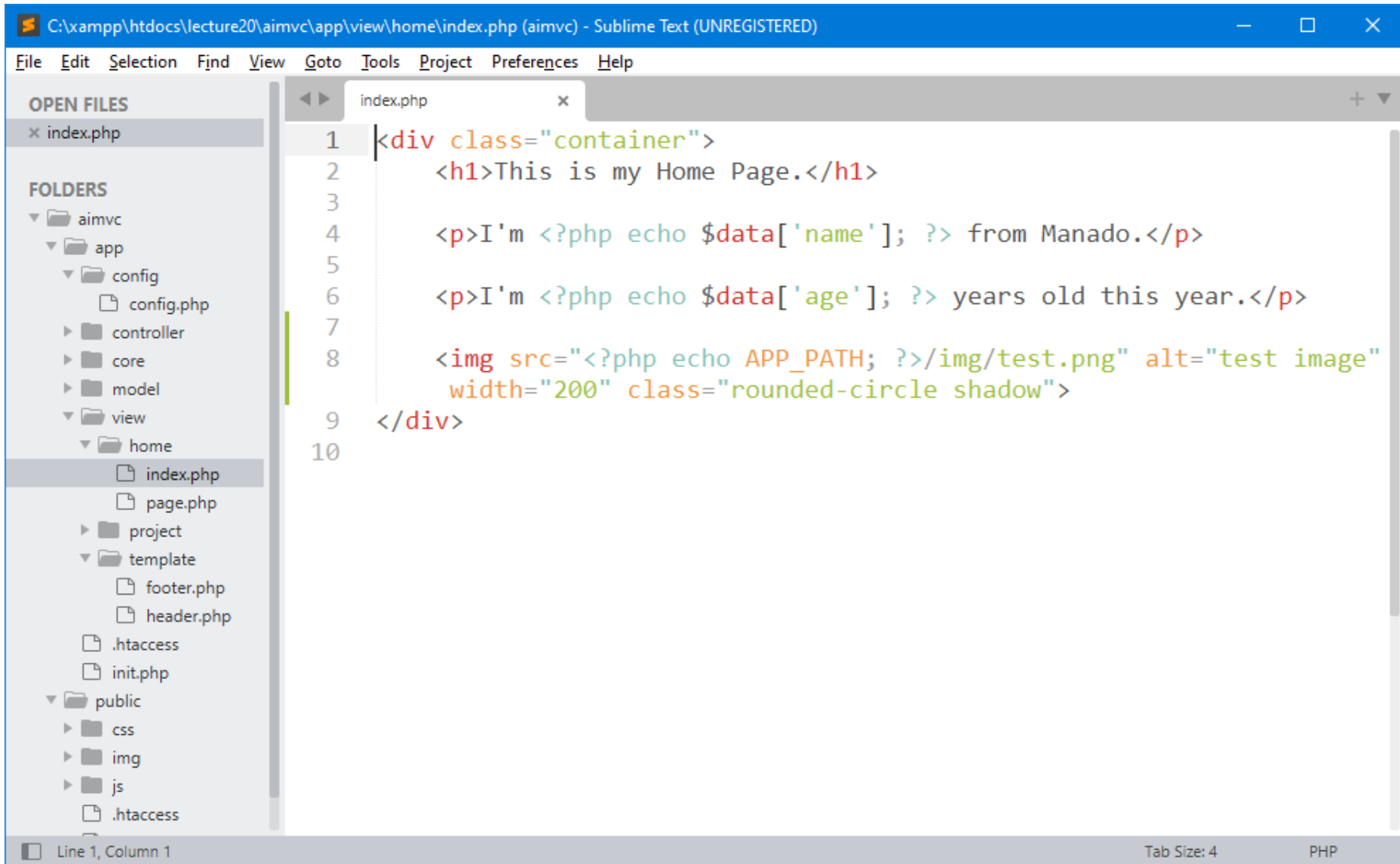
The slide features a white background with green leaves framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting they might be from a tree or shrub. They are arranged in a way that creates a natural border around the central text.

6 – Add Image

1 – Add Image in *img* directory



2 – Add *img* Tag in Script

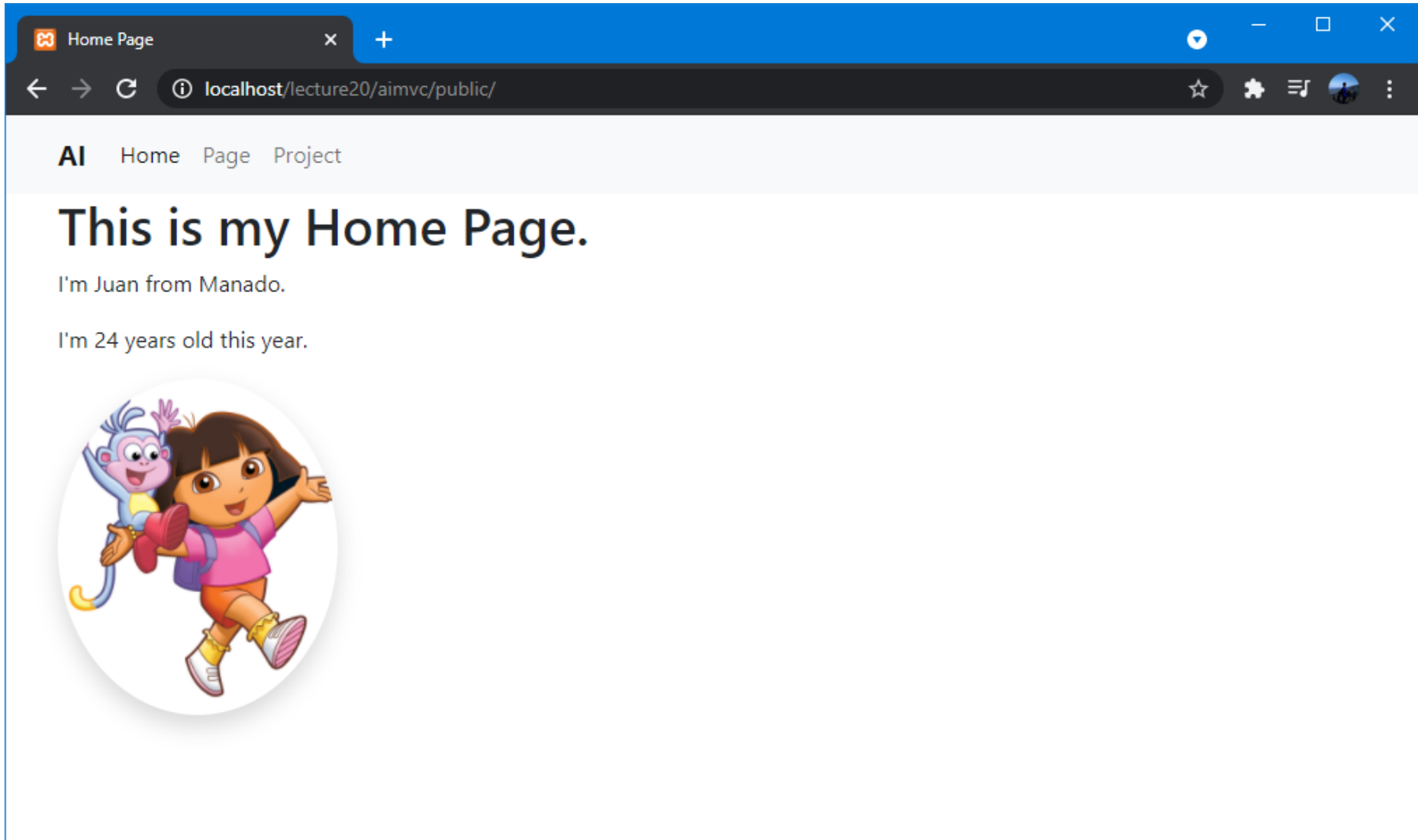


The screenshot shows the Sublime Text editor interface. The title bar indicates the file path is C:\xampp\htdocs\lecture20\aimvc\app\view\home\index.php (aimvc) - Sublime Text (UNREGISTERED). The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. The left sidebar shows the 'OPEN FILES' panel with 'index.php' selected, and the 'FOLDERS' panel showing the project structure. The main editor area displays the following PHP code:

```
1 <div class="container">
2   <h1>This is my Home Page.</h1>
3
4   <p>I'm <?php echo $data['name']; ?> from Manado.</p>
5
6   <p>I'm <?php echo $data['age']; ?> years old this year.</p>
7
8   
10 </div>
```

The status bar at the bottom shows 'Line 1, Column 1', 'Tab Size: 4', and 'PHP'.

3 – Testing UI Backend Framework

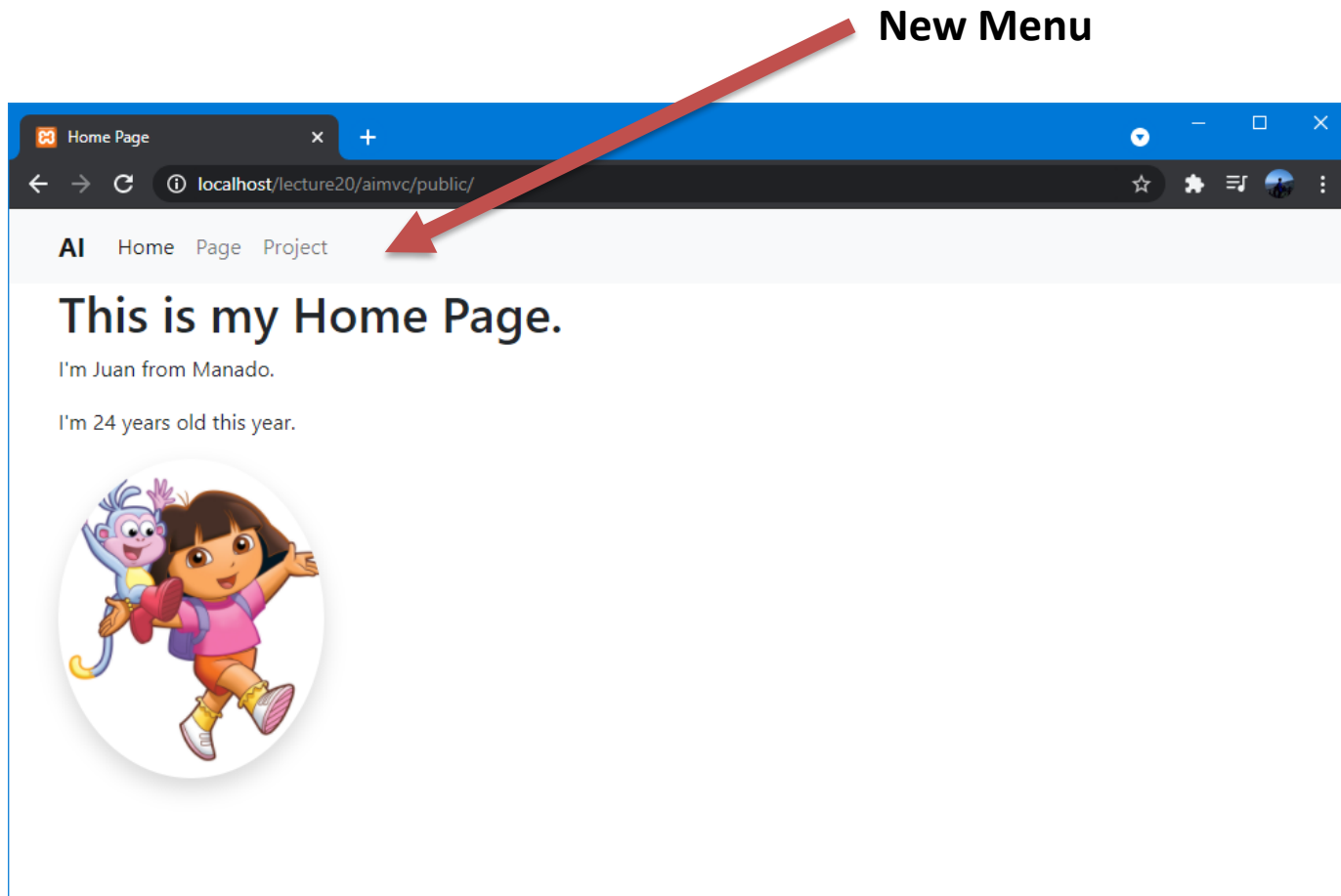




Exercise *for* Students

[PERTAMA]

**Lakukan uji coba di
local PC masing-
masing.**



Tambahkan sebuah menu (“**News**”) dan page yang baru di **controller Project**. Content untuk page pada menu News tersebut akan berisi *personal information* dari kalian (nama, alamat, umur, fakultas, jurusan, tingkat, agama, hobby, cita-cita, favorite quote, etc.).

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture20\
Create file : `_.php`

Buat semua contoh yang sudah di pelajari sebelumnya di *lecture* ini.

[SELANJUTNYA]

**Upload code Lecture #24
ke sisi server untuk di
DEMO.**

Create Directory *in* Server

[1] Masuk sebagai *administrator* atau *root*:

➤ ***sudo su***

[2] Buat project folder baru dengan perintah (Parent directory):

➤ ***sudo mkdir -p /var/www/html/lecture24***

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

➤ ***sudo chmod -R 755 /var/www/html/lecture24***

➤ ***sudo chown -R semmy:semmy /var/www/html/lecture24***

[4] Restart Apache Web Server:

➤ ***sudo systemctl restart apache2***

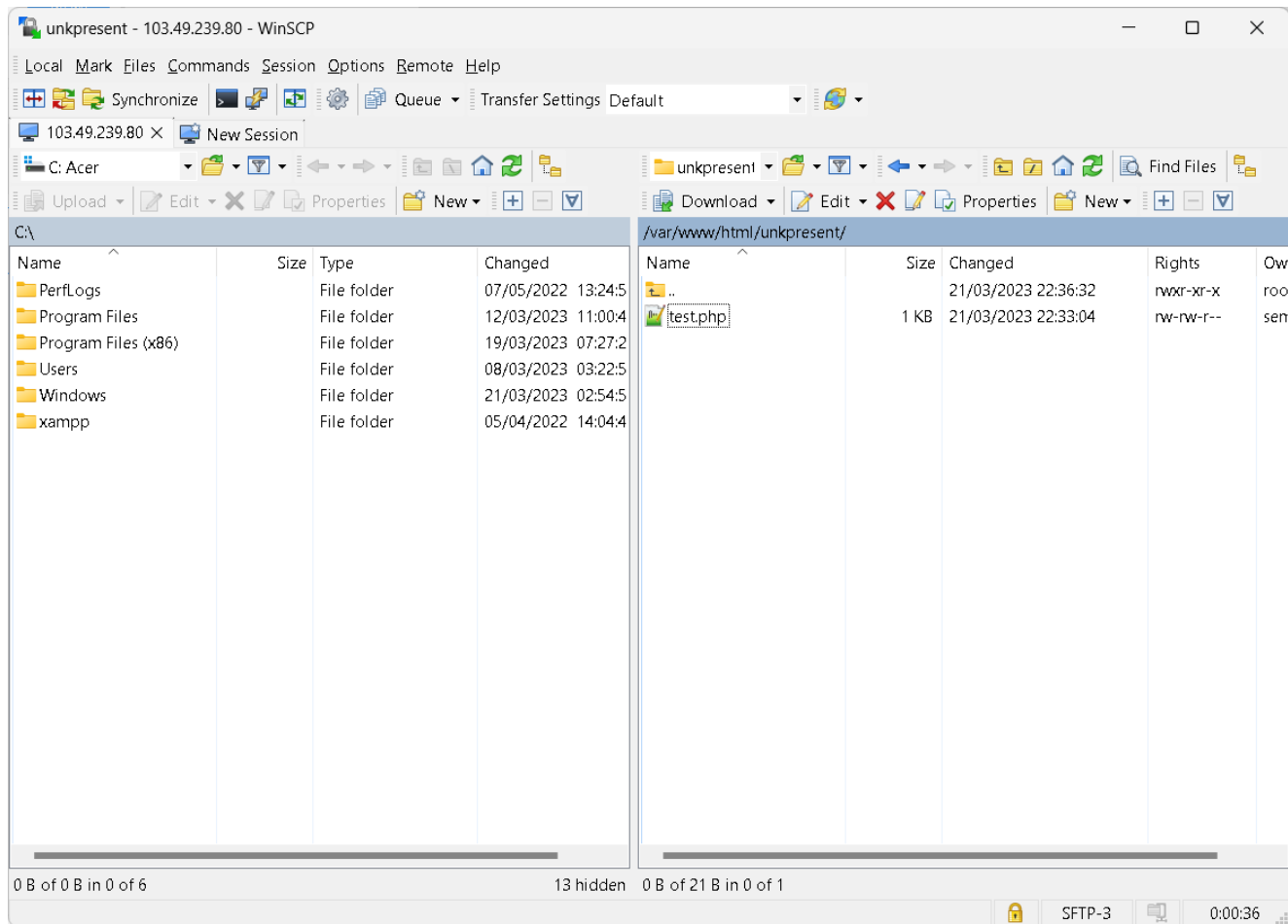
Note:

Ubah nama user “**semmy**” dengan nama *user* access ubuntu kalian.

```
root@unklab: /var/www/html
root@unklab:/home/semmy# sudo mkdir -p /var/www/html/unkpresent
root@unklab:/home/semmy# sudo chmod -R 755 /var/www/html/unkpresent
root@unklab:/home/semmy# cd ..
root@unklab:/home# cd ..
root@unklab:/# ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  tmp  var
boot  etc  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr
root@unklab:/# cd var
root@unklab:/var# ls
backups  crash  local  log  opt  snap  tmp
cache  lib  lock  mail  run  spool  www
root@unklab:/var# cd www/
root@unklab:/var/www# ls
html
root@unklab:/var/www# cd html/
root@unklab:/var/www/html# ls
index.html  test.php  unkpresent
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 root root 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html# sudo chown -R semmy:semmy /var/www/html/unkpresent
root@unklab:/var/www/html# sudo systemctl restart apache2
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 semmy semmy 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html#
```

Upload Project via WinSCP

- 1) Buka WinSCP dan arahkan ke “/var/www/html/lecture24”.
- 2) Selanjutnya, upload **file code lecture#24** dari local PC dan ke directory server menggunakan WinSCP.
- 3) Buka via Browser.



Output ditampilkan menggunakan domain name, HTTPS dan bisa di akses secara public via Browser:

Contohnya:

<https://domainkelompok.site/lecture24/aimvc/public/>

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

